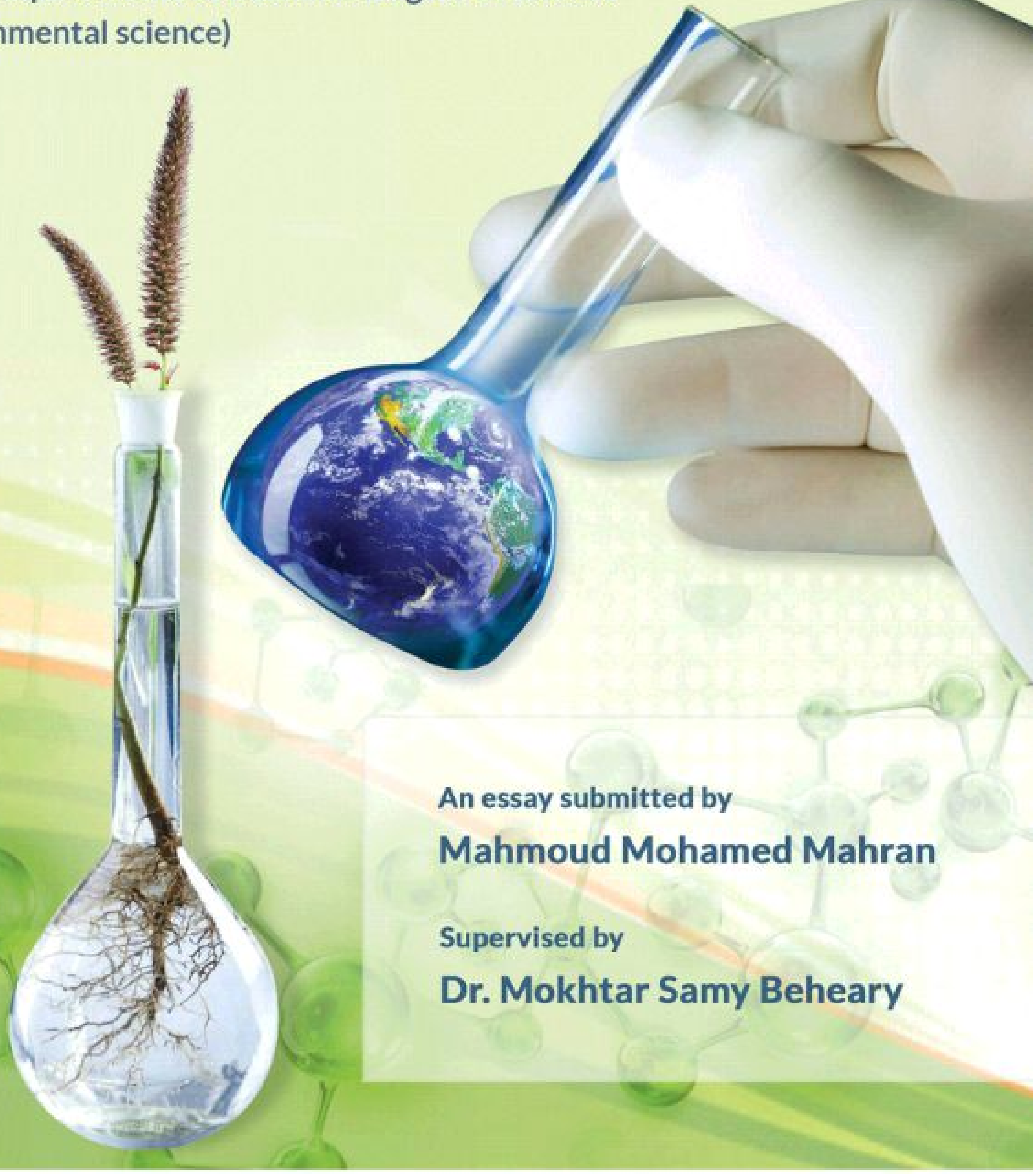
Portsaid University
Faculty of Science

Biological treatment of wastewater using plant cenchrus ciliaris

An essay submitted in partial fulfillment
for the requirements of Bachelor Degree of science
(Environmental science)



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Chapter 1

Introduction

Introduction

1.1 Water Pollution

Water pollution can be defined in many ways. Usually, it means one or more substances have built up in water to such an extent that they cause problems for animals or people. Oceans, lakes, rivers, and other inland waters can naturally clean up a certain amount of pollution by dispersing it harmlessly.

If you poured a cup of black ink into a river, the ink would quickly disappear into the river's much larger volume of clean water. The ink would still be there in the river, but in such a low concentration that you would not be able to see it.

At such low levels, the chemicals in the ink probably would not present any real problem. However, if you poured gallons of ink into a river every few seconds through a pipe, the river would quickly turn black.

The chemicals in the ink could very quickly have an effect on the quality of the water. This, in turn, could affect the health of all the plants, animals, and humans whose lives depend on the river.

Thus, water pollution is all about quantities: how much of a polluting substance is released and how big a volume of water it is released into. A small quantity of a toxic chemical may have little impact if it is spilled into the ocean from a ship.

But the same amount of the same chemical can have a much bigger impact pumped into a lake or river, where there is less clean water to disperse it.

Water pollution almost always means that some damage has been done to an ocean, river, lake, or other water source. A 1969 United Nations report defined ocean pollution as:

"The introduction by man, directly or indirectly, of substances or energy into the marine environment (including estuaries) resulting in such deleterious effects as harm to living resources, hazards to human health, hindrance to marine activities, including fishing, impairment of quality for use of sea water and reduction of amenities.

Fortunately, Earth is forgiving and damage from water pollution is often reversible. (UN Joint Group of Experts on the Scientific.1972)

During the last two decades large scale environmental initiatives have taken place in Europe and the United States, these have resulted in strict environmental regulation Markers of water pollution show the presence of pollution sources.

These include herbicides indicative of agricultural runoff, fecal coliform bacteria that are characteristic of pollution from sewage, and pharmaceuticals, pharmaceutical metabolites, and even caffeine that show contamination by domestic wastewater.

Biomarkers of water pollution are organisms that live in or are closely associated with bodies of water and that provide evidence of pollution either from accumulation in the organism of water pollutants or their metabolites or from effects on the organism owing to pollutant exposure.

Fish are the most common bio indicators of water pollution, and fish lipid (fat) tissue is commonly analyzed for persistent organic water pollutants, which tend to become concentrated in lipid tissue.

The osprey, a large, widely distributed fish-eating bird at the top of the food chain, has proven to be a good water pollution indicator species through chemical and biochemical analysis of its feathers, eggs, blood, and organs along with observations of behavior, nesting habits, and populations.

Even though it appears to be in plentiful supply on the earth's surface, water is a rare and precious commodity, and only an infinitesimal part of the earth's water reserves (approximately 0.03%) constitutes the water resource which is available for human activities.

The growth of the world's population and industry has given rise to a constantly growing demand for water in proportion to the supply available, which remains constant.

Thus, it is necessary to minimize its consumption and it is also necessary to return it back to the environment with the minimum contamination load because of the limited capacity of self-purification, hence the importance ns on the industrial emissions for the chemical industry.

It has been necessary to invest in cleaner technologies and in treatments that are more effective. On the other hand, numerous chemical companies have installed effluent treatment systems to meet the recently elaborated regulations of the country in which they are settled or to meet the regulations of the countries with which they trade.

The chemical industry comprises the companies that produce industrial chemicals. Basic chemicals or "commodity chemicals" are a broad chemical category including pharmaceutical products, polymers, bulk petrochemicals and intermediates, other derivatives and basic industrials, inorganic/organic chemicals, and fertilizers.

The chemical industry is of importance in terms of its impact on the environment. Chemical industrial wastewaters usually contain organic and inorganic matter in varying concentrations.

Many materials in the chemical industry are toxic, mutagenic, carcinogenic or simply almost non-biodegradable. This means that the production wastewater also contains a wide range of substances that cannot be easily degraded.

For instance, surfactant and petroleum hydrocarbons, among others chemical products that are being used in chemical industry reduce performance efficiency of many treatment unit operations.

1.2 Water pollution sources.

There are many sources of water pollution: direct discharges by industry, sewage, leaking underground storage tanks, and landfills are some point sources. But a major source of pollution comes from sources that are harder to pinpoint, and, therefore, harder to control.

These nonpoint sources, as they are called, include run-off that contains fertilizers, herbicides, and pesticides from agricultural lands and drainage from paved surfaces and storm drains.

Another source is not quite as obvious, but is deadly serious. Pollution from the air falls to the water as acid rain, killing lakes and causing harmful substances such as mercury to accumulate in fish.

Sources of water pollution are divided into two categories

- point source, or
- nonpoint source

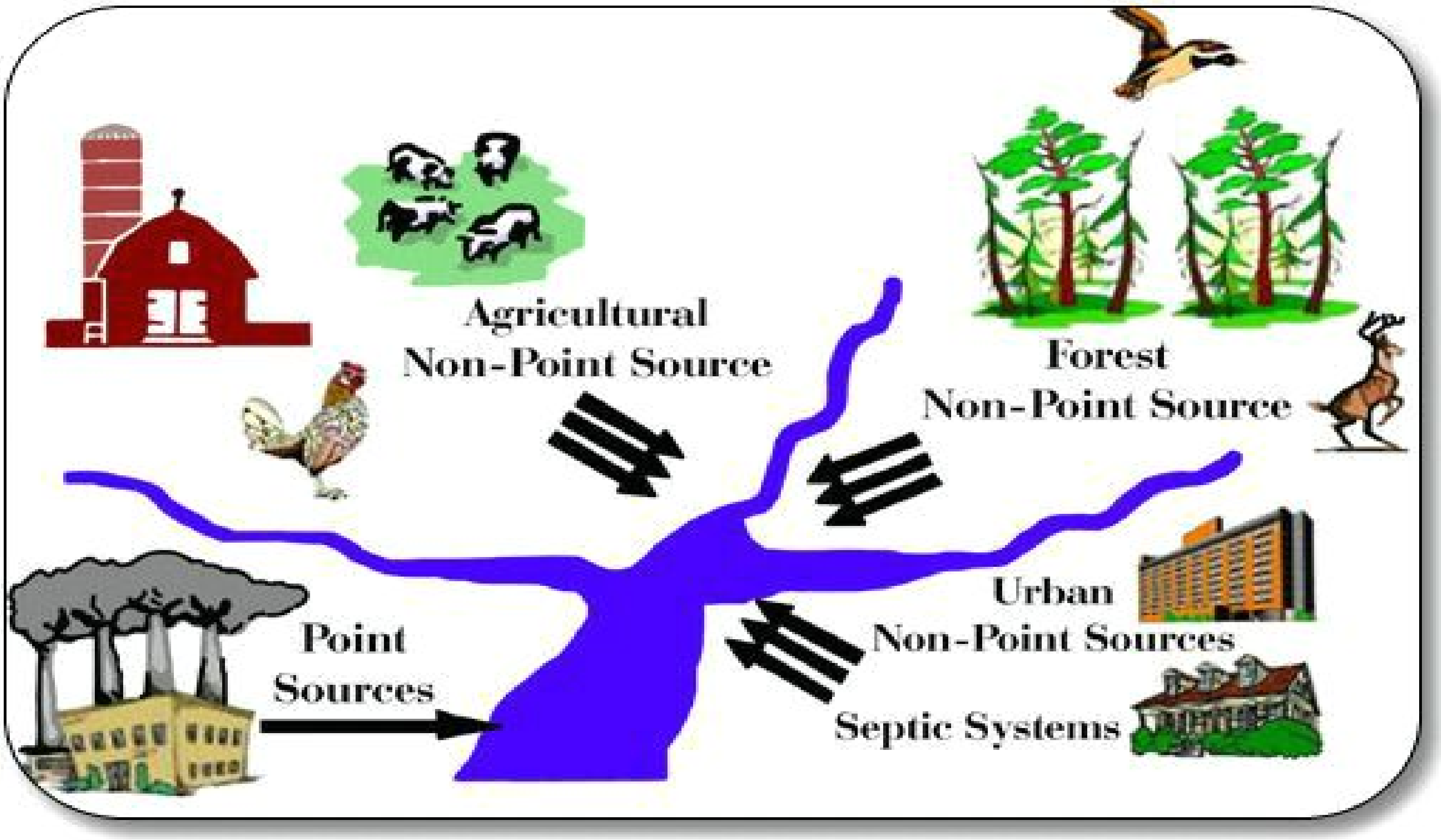

Fig 1.1 Sources of pollution to water bodies

1.2.1 Point source Water pollutant

A "point source" is "any single identifiable source. From which pollutants are discharged, e.g., a pipe, ditch, ship, or factory smokestack." Outlet pipes of industrial facilities or wastewater-treatment plants. The range of pollutants is vast, depending only on what gets "thrown down the drain."

1.2.1.1 Oxygen-demanding substances

such as might be discharged from milk processing plants, breweries, or paper mills, as well as municipal wastewater treatment plants, make up one of the most important types of pollutant because these materials decompose in the watercourse and can deplete the water's oxygen and create anaerobic conditions.

Oxygen-demanding substances measurement:-

Oxygen-demanding substances measured by Biochemical oxygen demand, where, Microorganisms decompose organic matter discharged to a water body. They require oxygen to do so.

The amount of oxygen required to decompose a given amount of organic pollutant is the "biochemical oxygen demand" (BOD). Natural BOD, such as plant debris and wildlife feces, is almost always present. However, a high BOD often indicates human activity, such as sewage or industrial discharge. Human activities that lead to a discharge of BOD include municipal wastewater treatment plants, food-processing operations, chemical plants, pulp and paper operations, tanneries, and slaughterhouses. (Baldry et al., 1991)

A high BOD can reduce or deplete the oxygen in water. In a large water body, fish can swim away from low-oxygen ("hypoxic") conditions, but crabs and snails and sedentary organisms may die. Hypoxia can be an irregular or seasonal occurrence in a water body, or an ongoing problem.

Can result. In estuaries and coastal areas, excessive nitrate is the major culprit although phosphorus an inorganic nutrient is not BOD, but a nutrient can generate BOD. Nitrate and phosphorus are two nutrients found in commercial fertilizer.

As inorganic chemicals, they themselves do not exert BOD. However, they are fertilizers, stimulating plant growth. Nutrients stimulate algal growth in water, sometimes an algal bloom. Zooplankton eat the proliferating algae. Bacteria, using the oxygen in water to do so, digest the fecal pellets of the zooplankton as well as dead plankton and dead vegetation.

Even if oxygen is only partially depleted, aquatic organisms suffer. If most of the oxygen is depleted, a dead zone, another important nutrient, also contributes to the problem. In freshwater lakes, phosphorus is often the major culprit stimulating excessive growth of vegetation. (Fennel, 2014)

1.2.1.2 Suspended solids

Also contribute to oxygen depletion; in addition, they create unsightly conditions and can cause unpleasant odors. This physical pollutant is found naturally in water to varying extents.

As usual, it is an excess that is deleterious. Also recall that in air, it is the very fine particles which cause the greatest health problems. Similarly, fine particles in soil runoff become fine suspended solids in water, which can cause serious problems.

Increased suspended solids content makes water more turbid or cloudy. This limits the sunlight reaching aquatic plants and stunts their growth. Fine suspended solids can also clog fish gills and harm the respiration of other water animals.

Suspended solids can interfere with efficient water disinfection by shielding microorganisms from the disinfectant. Surviving microorganisms can then contaminate drinking water. A major source of suspended solids is soil runoff from agricultural fields, especially in row crops. Forestry and construction activities contribute too. Point sources of suspended solids are facilities that discharge various solids including those that create BOD.

1.2.1.3 Nutrients

Mainly nitrogen and phosphorus, can promote accelerated. A nutrient is a substance required for life, but has a more sinister face at high concentrations. Man-made fertilizers, containing concentrated reactive nitrogen and phosphorus are a major source of nutrients to water bodies.

When fertilizer is added to agricultural fields, the excess runs off with rainwater into water bodies. There water plants, especially algae, ingest the nutrients. This has a cascade of effects. Although synthetic fertilizer is the major offender, any organic matter has nutrient value.

Human activities that lead to the discharge of organic matter include municipal wastewater-treatment plants, food-processing operations, chemical plants, pulp and paper operations, tanneries, and slaughterhouses.

Natural sources of nutrients likewise are similar to those noted for BOD, i.e., plant and animal debris and wildlife feces. Ebbers, B., Ottosen, L. M. & Jensen, P. E. (2015).

Continuing input of excess nutrients can lead to eutrophication, a process " during which a lake, estuary, or bay evolves into a bog or marsh and eventually disappears. "A water body naturally becomes eutrophic, but over many years as it slowly accumulates nutrients.

During later stages of eutrophication, the water is choked with plant life, in particular algal "blooms." Blooms may form a scum on the water surface, produce offensive smells, give the water a bad taste, and make it unfit for swimming.

Human activities that put excess amounts of nutrients into water accelerate eutrophication.3 Nitrate and ammonia, as well as many organic chemicals, contain nitrogen in a form bioavailable to plants and algae. Excess nutrients the "nitrogen glut"-- have become a global problem. Chen, Y., Randall, A. A. & Mccue, T. (2004)

1.2.1.4 PH

Acid deposition. The global problem of acid deposition can lead to water bodies becoming too acid to optimally support life or, sometimes, to support life at all. Mining operations. These are another source of damaging amounts of acid. Metal ores often contain metal sulfides.

Mining of sulfide ores and of sulfur-containing coal brings sulfides to Earth's surface. There, exposed to oxygen, sulfides are oxidized to sulfate, and in the presence of moisture sulfate is converted to sulfuric acid. This acid runs off into nearby water, sometimes causing great damage.

To make the situation even worse, sulfuric acid dissolves metals, including hazardous metals, contained in mining wastes; rainwater runoff can carry these into water bodies.

The acid damage can sometimes be carried for miles downstream. If iron is one of the metals present, contaminated streams can turn orange. G. Allen Burton, Jr., Robert Pitt (2001)

1.2.1.5 Bio concentrated metals

Can adversely affect aquatic ecosystems as well as make the water unusable for human contact or consumption. Metals are naturally found to varying extents in food and drinking water. Indeed nutrient metals (iron, chromium, copper, cobalt, manganese, molybdenum, nickel, selenium, vanadium, and zinc) are needed.

However, at higher doses, metals can be toxic think of iron, a nutrient. Iron still occasionally kills small children who find and eat their parents' iron supplement capsules.

Natural metal levels can also pose risks when levels are unusually high; arsenic is found naturally at high levels in some soils and waters. However, human activities are responsible for the dramatic increases in metals in the environment.

That century saw the mining of about 90% of all cadmium, copper, lead, nickel, and zinc ever mined.

The metals, lead, cadmium, and mercury pose special concerns. Others that sometimes present problems include antimony, cobalt, copper, manganese, molybdenum, nickel, selenium, vanadium, and zinc. G. Allen Burton, Jr., Robert Pitt (2001)

1.2.1.6 Heat

Is also an industrial waste that is discharged into water; heated discharges may drastically alter the ecology of a stream or lake.

Although local heating can have beneficial effects such as freeing harbors from ice, the primary effect is deleterious:

Lowering the solubility of oxygen in the water, because gas solubility in water is inversely proportional to temperature, and thereby reducing the amount of dissolved oxygen (DO) available to gill-breathing species. As the level of DO decreases, metabolic activity of aerobic aquatic species increases, thus increasing oxygen demand. Fennel, K. (2014)

1.2.1.7 Municipal wastewater

Is as important a source of water pollution as industrial waste. A century ago, most discharges from municipalities received no treatment whatsoever. Since that time, the population and the pollution contributed by municipal discharge have both increased, but treatment has increased also.

We define a population equivalent of municipal discharge as equivalent to the amount of untreated discharge contributed by a given number of people. Ebbers, B., Ottosen, L. M. & Jensen, P. E. (2015)

1.2.1.8 Agricultural wastes

Should they flow directly into surface waters, have a collective population equivalent of about 2 billion. Feedlots where large numbers of animals are penned in relatively small spaces provide an efficient way to raise animals for food.

They are usually located near slaughterhouses and thus near cities. Feedlot drainage (and drainage from intensive poultry cultivation) creates an extremely high potential for water pollution. Aquaculture has a similar problem because wastes are concentrated in a relatively small space.

1.2.1.9 Sediment from land erosion

May also be classified as a pollutant. Sediment consists of mostly inorganic material washed into a stream as a result of land cultivation, construction, demolition, and mining operations.

Sediment interferes with fish spawning because it can cover gravel beds and block light penetration, making food harder to find. Sediment can also damage gill structures directly. Nashville, TN (2012)

1.2.1.10 Pollution from petroleum compounds ("oil &grease pollution")

Oil spills are a major problem in some near-coastal waters, killing or adversely affecting fish, other aquatic organisms, birds, and mammals. These spills can also kill or reduce organisms living in coastal sands and rocks, and may kill the worms and insects that are food to birds and wildlife.

When the spills intrude into coastal marshes, the oil can damage or kill fish, shrimp, birds, and other animals. Oil spills can also foul beaches used for swimming and recreation.

Despite the sometimes horrendous damage caused by oil spills, they are seen as a relatively minor problem for fish and the marine environment in comparison to chronic nutrient pollution.

Depending upon the amount and type of oil spilled, where it is spilled, and weather conditions, ecosystem recovery can be quick or painfully slow. Spills are not the only source of oil in water: oil leaking from vehicles, or released during accidents, washes off roads with rainwater and then reaches water bodies.

A portion also percolates down to groundwater. Used oil from motor vehicles is often improperly disposed of too. Direct releases of oil into water bodies also occur.

Motor and other recreational boats release up to 30% of their fuel, unburned, into water. These individually small, but ongoing events add up to much more oil than is spilled in a dramatic event such as the Exxon Valdez.

However, the effects of a major spill are obvious whereas the environmental impact of ongoing small events is harder to assess.G. Allen Burton, Jr., Robert Pitt (2001)

1.2.1.11 Pathogenic microorganisms

Most microorganisms are not pathogens, and do not cause disease. Most perform useful, often vital, functions for humans including assisting the digestion in our intestines. We depend on microbes to degrade organic wastes in the environment and to biodegrade the organic material in landfills.

We use microbes in fermentations to make food products, pharmaceuticals, and other chemicals. Microbes are almost ubiquitous in our environment, found almost anywhere that one looks.

Nevertheless, we rarely get sick much less succumb to these infections. "It is specific microorganisms, pathogens that are capable of causing infectious disease, that concern us.

A pathogen can be a bacterium, virus, fungus, protozoan, or toxic algal species. If infectious microbes or their toxins are found in shellfish, their harvesting for food use is halted.

Pathogenic viruses and bacteria in coastal water can infect swimmers and others. Viruses are abundant in marine waters, often surviving in salt water longer than bacteria. Infections can result not just by ingesting water containing Pathogens, but through the skin. Harrison, Roy M., ed. (2001)

Pathogenic microbes in a water body are often anthropogenic, generated by human activities runoff. They may arrive in water bodies in runoff of storm water, and from improperly operating septic systems, or runoff from livestock operations.

Pathogens sometimes come from point sources, especially poorly performing municipal sewage-treatment plants. All these sources exist in developed countries. The situation is worse in less-developed nations where most sewage remains untreated, and is often dumped into rivers and oceans.

1.2.2 Nonpoint source pollution

A "non-point-source" pollutant is one whose source is much harder to identify precisely, hence the term "non-point". "Runoff" indicates rainwater or snowmelt carried across land to water.

Runoff arises from non-point sources. Runoff carries almost anything that water can carry - oil, grease, dirt, trash, animal waste, microorganisms, and chemical pollutants, including metals, pesticides, and fertilizers. Urban non-point sources include streets and parking lots, roofs, and construction sites. Rural non-point sources include agriculture, logging, and mining sites. Pollution from non-point sources is much harder to control than that from point sources.

On undisturbed land, rainwater can percolate down through the soil to replenish groundwater. As the water moves downward, the soil providing one of nature's services absorbs and detoxifies many pollutants.

But on land covered with parking lots, roads, shopping malls, factories, buildings, and homes there is less soil to absorb water. In addition to less replenishment of groundwater, contaminant-carrying rainwater runs off into surface water. Moss, Brian (2008)

1.2.2.1 Reducing non-point sources

Reducing non-point-source pollution is much more difficult than reducing point -source pollution. Table 1 shows sources of polluted rainwater and snowmelt runoff, the contaminants, and ways to reduce them.

The readily identified by effluent pipes are fairly tightly controlled." The problem now "is in large part the result of runoff from roads and urban areas, farms, and timber operations.

1.3 Impact Of Water Pollution

1.3.1 Impact of Water Pollution on Human Health

- According to the UN World Water Assessment Program, about 2.3 billion people suffer from diseases associated with polluted water, and more than 5 million people die from these illnesses each year.
- Other illnesses—such as malaria, filariasis, yellow fever, and sleeping sickness—are transmitted by vector organisms (such as mosquitoes and tsetse flies) that breed in or live near stagnant, unclean water.
- A number of chemical contaminants—including DDT, dioxins, polychlorinated biphenyls Dysentery, typhoid, cholera, and hepatitis A are some of the ailments that result from ingesting water contaminated with harmful microbes.
- (PCBs), and heavy metals— are associated with conditions ranging from skin rashes to various cancers and birth defects.
- Excess nitrate in an infant's drinking water can lead to the "blue baby syndrome" (methemoglobinemia) a condition in which the child's digestive system cannot process the nitrate, diminishing the blood's ability to carry adequate concentrations of oxygen.

1.3.2 Effects of Water Pollution on animals

- Nutrient pollution (nitrogen, phosphates etc.) causes overgrowth of toxic algae eaten by other aquatic animals, and may cause death; nutrient pollution can also cause outbreaks of fish diseases.
-
- Chemical contamination can cause declines in frog biodiversity and tadpole mass Oil pollution (as part of chemical contamination) can negatively affect development of marine organisms, increase susceptibility to disease and affect reproductive processes; can also cause gastrointestinal irritation, liver and kidney damage, and damage to the nervous system

- Mercury in water can cause abnormal behavior, slower growth and development, reduced reproduction, and death
- Persistent organic pollutants (POPs) may cause declines, deformities and death of fish life
- Too much sodium chloride (ordinary salt) in water may kill animals.

1.3.3 Impact Of water pollution on ecosystems

- Water pollution has adverse effects on ecosystems For instance, while moderate amounts of nutrients in surface water are generally not problematic, large quantities of phosphorus and nitrogen compounds can lead to excessive growth of algae and other nuisance species.
- Known as eutrophication, this phenomenon reduces the penetration of sunlight through the water; when the plants die and decompose, the body of water is left with odors, bad taste, and reduced levels of dissolved oxygen.
- Low levels of dissolved oxygen can kill fish and shellfish. In addition, aquatic weeds can interfere with recreational activities (such as boating and swimming) and can clog intake by industry and municipal systems.
- Some pollutants settle to the bottom of streams, lakes, and harbors, where they may remain for many years. For instance, although DDT and PCBs were banned years ago, they are still found in sediments in many urban and rural streams. They occur at levels harmful to wildlife at more than two-thirds of the urban sites tested. (Beheary, 2013).

1.3.4 Effects of Water Pollution on plants

• May disrupt photosynthesis in aquatic plants and thus affecting ecosystems that depend on these plants

• Terrestrial and aquatic plants may absorb pollutants from water (as their main nutrient source) and pass them up the food chain to consumer animals and humans

• Plants may be killed by too much sodium chloride (ordinary slat) in water

• Plants may be killed by mud from construction sites as well as bits of wood and leaves, clay and other similar materials

• Plants may be killed by herbicides in water; herbicides are chemicals which are most harmful to plants.

1.4 Traditional treatment of wastewater

In terms of wastewater treatment there are four classifications of treatment. Preliminary treatment involves the removal of large particles as well as solids found in the wastewater.

The second classification is primary treatment, which involves the removal of organic and inorganic solids by means of a physical process, and the effluent produced is termed primary effluent.

The third treatment is called secondary treatment; this is where suspended and residual organics and compounds are broken down. Secondary treatment involves biological (bacterial) degradation of undesired products. The fourth is tertiary treatment, normally a chemical process and very often including a residual disinfection.

1.4.1 Physico-chemical treatment

1.4.1.1 Coagulation –flocculation

Most wastewater treatment plant includes sedimentation in their process. The sedimentation also called clarification is a treatment process in which the velocity of the water is lowered below the suspension velocity and the suspended particles settle out of the water due to gravity.
Settled solids are removed as Sludge, and floating solids are removed as scum. Wastewater leaves the sedimentation tank over an effluent weir to the next step of treatment.

The efficiency or performance of the process is controlled by: retention time, temperature, tank design, and condition of the equipment. However, without coagulation/flocculation, sedimentation can remove only coarse suspended matter which will settle rapidly out of the water without the addition of chemicals.

This type of sedimentation typically takes place in a reservoir, sedimentation or clarification tank, at the beginning of the treatment process. Coagulation-flocculation consists on the addition on the clarification tanks of chemical products that accelerate the sedimentation (coagulants).

The coagulants are inorganic or organic compounds such as Aluminum sulphate, aluminum Hydroxide chloride or high molecular weight cationic polymer. The purpose of the addition of coagulant is to remove almost 90% of the suspended solids from the wastewater at this stage in the treatment process.

1.4.1.2 Adsorption techniques to treat wastewater

Adsorption is a natural process by which molecules of a dissolved compound collect on and adhere to the surface of an adsorbent solid. Adsorption occurs when the attractive forces at the carbon surface overcome the attractive forces of the liquid.
Granular activated carbon is a particularly good adsorbent medium due to its high surface area to volume ratio. One gram of a typical commercial activated carbon will have a surface area equivalent to 1,000 square meters.

1.4.1.3 Granular activated carbon

The pollution of water resources due to the indiscriminate disposal of heavy metals has been causing worldwide concern for the last few decades. It is well known that some metals can have toxic or harmful effects on many forms of life.

Metals, which are significantly toxic to human beings and ecological environments, include chromium (Cr), copper (Cu), lead (Pb), mercury (Hg), manganese (Mn), cadmium (Cd), nickel (Ni), zinc (Zn) and iron (Fe), etc. This problem has received considerable amount of attention in recent years. One primarily concern is that marine animals which can readily absorb those heavy metals in wastewater and directly enter the human food chains present a high health risk to consumers. Wastewater from many industries such as metallurgical, tannery, chemical manufacturing, mining, battery manufacturing industries, etc.

Contains one or more of these toxic heavy metals. Industries carries out operations like electroplating, metal/surface finishing and solid-state wafer processing, generate wastewater contaminated with hazardous heavy metals. (Perisamy et al (1996)

The concentrations of some of the toxic metals like Cr, Hg, Pb, As, etc. are higher than permissible discharge levels in these effluents. It, therefore, becomes necessary to remove these heavy metals from these wastewaters by an appropriate treatment before releasing them into the environment.
Tsai MJ, Lin LD (2012)

1.4.1.4 Electro sorption

Electro sorption is generally defined as potential polarization induced adsorption on the surface of electrodes, and is a non-Faraday process. After the polarization of the electrodes, the polar molecules or ions can be removed from the electrolyte solution by the imposed electric field and adsorbed onto the surface of the electrode.

Because of its low energy consumption and environmentally friendly advantage, electro sorption has attracted a wide interest in the adsorption processes for treatment of wastewater. Although electro sorption has been shown as a promising treatment process, it has been limited by the performance of electrode material.

Activated carbon fiber cloth with high specific surface area and high conductivity is one of the commonly used electrode materials. The surface chemistry of activated carbon fiber has been recognized as a key parameter in the control of the adsorption process.

To increase the adsorption capacity, a number of modification methods have been employed . The adsorption capacity and adsorption kinetics depend on the surface properties of adsorbent. To increase the potentially low adsorption capacity of any adsorbent, a number of modifications including immobilization of a chelating agent on the adsorbent surface have been employed. (Adhoum N, Moser L(2002)

The adsorption capacities and the feasible removal rates must be substantially boosted by the modification techniques. Ethylene diamine tetra acetic acid (EDTA) is the most widely used of the amino poly carboxylic acids. EDTA is a chelating agent, forming coordination compounds with most monovalent, divalent, trivalent and tetravalent metal ions.

It combines with metal ions in a 1:1 ratio regardless of the charge on the cation. Activated carbon cloth is known for its effectiveness in removing chemicals from water and wastewater.

Loading of C-cloth with EDTA provides a more efficient sorbent for the adsorption of metal cations. Modification of C-cloth with EDTA causes a significant increase in the rate and the extent of the adsorption of metal cations.

Procedures based on adsorption of some cations at EDTA loaded high-area C-cloth are shown to be effective for removal of them from aqueous solutions. Langmuir model is more successful than Freundlich model in representing experimental isotherm data for the adsorption of the most of the ions on both C-cloths. Afghani et al (2008) and Huang et al (2010)

1.4.2 Biological treatment of chemical industry wastewater

Aerobic treatment: In the wastewater treatment sector, biological processes deal primary with organic impurities. Microbial-based technologies have been used over the last century for the treatment of liquid waste domestic stream.

The development of these technologies has provided excellent process for the destruction of waste constituents that are readily biodegradable under aerobic conditions.

Therefore, processes similar to those used for conventional domestic wastewater treatment have applied successfully to the treatment of many industrial wastewaters.

Aerobic degradation in the presence of oxygen is considered to be a relatively simple, inexpensive and environmentally sound way to degrade wastes. Factors that are critical in the optimal degradation of the selected substrate include the temperature, moisture, pH, nutrients and aeration rate that the bacterial culture is exposed to, with temperature and aeration being two of the most critical parameters that determine the degradation rates by the microorganism. Soluble organic sources of biochemical oxygen demand (BOD) can be removed by any viable microbial process, aerobic, anaerobic or anoxic.

However, aerobic processes are typically used as the principal means of BOD reduction of domestic wastewater because the aerobic microbial reactions are fast, typically 10 times faster than anaerobic microbial reactions.

Therefore, aerobic reactors can be built relatively small and open to the atmosphere, yielding the most economical means of BOD reduction. On the other hand, the major disadvantage of aerobic bioprocesses for waste treatment, relative to anaerobic processes, is the large amount of sludge produce.

A relatively high accumulation of biomass occurs in the aerobic bioreactor because the biomass yield (mass of cell produced per unit mass of biodegradable organic matter) for aerobic microorganisms is relatively high, almost 4 times greater than the yield for anaerobic organisms.

The sludge present in the reactor effluent can contain residual BOD that may need to be reduced in an additional process, and must ultimately be disposed of as a solid waste.

There are many mechanisms that are utilized by the microorganisms during the aerobic degradation process. Some of these include the attack on the xenobiotics by organic acids produced by the microorganisms, the production of noxious compounds like hydrogen supplied and the production of chelating agents which are able to increase the solubility of any insoluble xenobiotics, making them more available to the microorganisms and mechanical degradation.

The wastewater from chemical industries may exert a toxic effect on the microorganisms present in conventional activated sludge reactor. Chemical compounds found in these wastewater streams cannot be utilized as a sole carbon source by microorganisms and will vary in toxicity.

Inhibition of growth of the microorganisms by these components therefore plays a crucial role in the degradation process, as this can cause the treatment system to fail Ren S et al (2004)&(WelanderU et al (2011)
The key to successful bioremediation technology of some chemical industries wastewater is to modify or optimise the cell/substrate contact time, so that biodegradation can proceed in a reasonable time and potential toxicity of the wastewater of the wastewater to the micro flora is reduced.

According to the literature, the best option for bioremediation of this type of wastewater is a membrane bioreactor (MBR) that has been inoculated with activated sludge, which has been shown to effectively treat high-strength organic wastes.

On the other hand, the two-phase partitioning reactor has also been effective with toxic substrates. The following section details two activated aerobic sludge systems that have been shown to facilitate degradation of xenobiotics in the presence of toxic compounds. (Malinowski JJ (2001)

1.4.3 Chemical oxidation

Oxidation, by definition, is a process by which electrons are transferred from one substance to another. This leads to a potential expressed in volts referred to a normalized hydrogen electrode.

From this, oxidation potentials of the different compounds are obtained. Chemical oxidation appears to be one of the solutions to be able to comply with the legislation with respect to discharge in a determined receptor medium. It can also be considered as an economically viable previous stage to a secondary treatment of biological oxidation for the destruction of non-biodegradable compounds, which inhibit the process.

A reference parameter in case of using chemical oxidation as treatment process is the chemical oxygen demand (COD). Only waters with relatively small COD contents (≤5 g.L-1) can be suitably treated by means of these processes since higher COD contents would require the consumption of too large amounts of expensive reactants.

In those cases, it would be more convenient to use wet oxidation or incineration: waste water with COD higher than 20 g.L-1 may undergo auto thermic wet oxidation [35]. The chemical oxidation processes can be divided in two classes:

1.4.3.1 Classical Chemical Treatments

Advanced Oxidation Processes (AOPs) Classical chemical treatment: Classical chemical treatments consist generally on the addition of an oxidant agent to the water containing the contaminant to oxidize it.

Among the most widely used it is possible to emphasize the following classical oxidants. - Chlorine: it is a good chemical oxidizer for water evaporation because it destroys microorganisms. It is a strong and cheap oxidant, very simple to feed into the system and it is well known. (Rodrıǵuez M (2003)

Its main disadvantages are its little selectivity that high amounts of chlorine are required and it usually produces carcinogenic organo chloride byproducts. - Potassium permanganate: It has been an oxidizer extensively used in the treatment of water for decades.

It can be introduced into the system as a solid or as a solution prepared on site. It is a strong but expensive oxidant, which works properly in a wide pH range. One of the disadvantages of the use of potassium permanganate as an oxidizer is the formation of magnesium dioxide throughout oxidation, which precipitate and has to be eliminated afterward by clarifying or filtration, both of which mean an extra cost.

1.4.3.1.1 Oxygen

The reaction of organic compounds with oxygen does not take place in normal temperature and pressure conditions. Needed values of temperature and pressure are high to increase the oxidizing character of the oxygen in the reaction medium and to assure the liquid state of the effluent.

It is a mild oxidant that requires large investments in installations. However, its low operating costs make the process attractive.

1.4.3.1.2 Hydrogen peroxide

It is a multipurpose oxidant for many systems. It can be applied directly or with a catalyst. The catalyst normally used is ferrous sulphate (the so-called Fenton process, which will be presented below). Other iron salts can be used as well. Other metals can also be used as catalyst, for example, Al3+, Cu2+. Its basic advantages are: (i) it is one of the cheapest oxidizers that is normally used in residual, (ii) waters, (iii) it has high oxidizing power, (iv) it is easy to handle, (v) it is water-soluble, (vi) it does not produce toxins or color in by products.

It can also been used in presence of ultraviolet radiation and the oxidation is based on the generation of hydroxyl radicals that will be considered an advanced oxidation process.

An option to the ending of hydrogen peroxide to the reaction medium is its production on site. One production possibility is by electro reduction of the oxygen dissolved in the reaction medium. (Rodríguez M (2003)

This option is not used very much, because it is expensive and increases the complexity of the system.

1.4.3.1.3 OZonation

It is a strong oxidant that presents the advantage of, as hydrogen peroxide and oxygen, not introducing "strange ions" in the medium. Ozone is effective in many applications, like the elimination of color, disinfection, elimination of smell and taste, elimination of magnesium and organic compounds.

In standard conditions of temperature and pressure it has a low solubility in water and is unstable. It has an average life of a few minutes. (Rodríguez M (2003)

Therefore, to have the necessary quantity of ozone in the reaction medium a greater quantity has to be used. Among the most common oxidizing agents, it is only surpassed in oxidant power by fluorine and hydroxyl radicals.

Although included among the classical chemical treatments, the oZonation of dissolved compounds in water can constitute as well an AOP by itself, as hydroxyl radicals are generated from the decomposition of ozone, which is catalyzed by the hydroxyl ion or initiated by the presence of traces of other substances, like transition metal cations.

As the pH increases, so does the rate of decomposition of ozone in water. The major disadvantage of this oxidizer is that it has to be produced on site and needs installation in an ozone production system in the place of use.

Therefore, the cost of this oxidizer is extremely high, and it must bear this in mind when deciding the most appropriate oxidizer for a given system. In addition, as it is a gas, a recuperation system has to be foreseen and that will make the obtaining system even more expensive.

OZonation is used in many drinking water plants as a tertiary treatment and also for the oxidation of organic pollutants of industrial (paper mill industry) or agriculture (water polluted by pesticides) effluents.

1.4.3.2 Advanced Oxidation Processes (AOPs)

AOPs were defined by Glaze and Chapin. (Glaze W, Chapin D (1987) as near ambient temperature and pressure water treatment processes which involve the generation of highly reactive radicals (specially hydroxyl radicals) in sufficient quantity to effect water purification.

These treatment processes are considered as very promising methods for the remediation of contaminated ground, surface, and wastewaters containing non -biodegradable organic pollutants. Hydroxyl radicals are extraordinarily reactive species that attack most of the organic molecules.

(Merayo et.al (2013)

Chapter2

"*Cenchrus ciliaris*" plant

Plant *Cenchrus ciliaris*

2.1 characteristics

The global cost of controlling biological invasions is phenomenal. Typically, mitigation, containment or eradication of the invasive species is desired; however, control and eradication become controversial if the species is economically significant.

Buffel grass (*Cenchrus ciliaris* L.) is grown widely in tropical and sub-tropical arid rangelands around the globe because of its high tolerance to drought and capacity to withstand heavy grazing.

Outside its natural range, Buffel grass can rapidly invade native vegetation, roadsides and urban landscapes, altering the wildfire regime and displacing the native flora and fauna.

Due to the economic benefits of the species, eradication is controversial and weed management authorities are ill-informed to effectively target management actions.

While over 400 research papers have been published relating to the improvement of Buffel pasture, less than 20 relate to its impact on biodiversity and even fewer describe its nature as an invader (Web of Science, June 2011).

Strategic control of Buffel grass invasions requires knowledge of regions infested with or vulnerable to invasion, as well as a willingness from the community to be involved in controlling its spread, all of which are currently lacking.
Presented here is a review of the ecology, distribution and biodiversity impacts of Buffel grass in invaded environments, as well

As a synthesis of physiological characteristics relevant to an understanding its behaviour as an invader.

The paper aims to increase awareness about the ecological dangers of Buffel grass invasions continuing unchecked and to improve understanding about the ecology of Buffel grass for the purpose of managing invasions.

2.2 Identity

Preferred Scientific Name: *Cenchrus ciliaris* L.
Preferred Common Name: Buffel grass
Other Scientific Names: Cenchrus bulbosus Fresen.
International Common Names
- **English:** African foxtail grass; African foxtailgrass; buffelgrass
- **Spanish:** yerba de salinas; Zacate buffel
- **French:** cenchrus cilié; cenchrus de Rhodésie

Local Common Names
- **Brazil:** capim-búfel
- **Cuba:** guisaso
- **Italy:** cencro ciliare
- **Mexico:** pasto buffel; zacate buffel
- **Puerto Rico:** yerba de salinas
- **India:** anjan; anjan grass; dhaman

South Africa: Bloubuffelsgras

EPPO code

PESCI (Pennisetum ciliare)

Taxonomic Tree

Domain: Eukaryota
Kingdom: Plantae
Phylum: Spermatophyta
Subphylum: Angiospermae
Class: Monocotyledonae
Order: Cyperales
Family: Poaceae
Genus: Pennisetum
Species: *Cenchrus ciliaris*

2.3 Nomenclature and morphology

Buffel grass (*Cenchrus ciliaris* L.) is a robust, deep-rooted C4 perennial tussock grass native to tropical and sub-tropical arid environments (Sharif-Zadeh and Murdoch, 2001).

The species has highly varied morphological and physiological characteristics, which when combined with its wide geographic distribution, have led to considerable taxonomic uncertainty, with many synonyms evolving as a result. The common name Buffel grass is widely accepted and generally refers to *Cenchrus ciliaris* or Pennisetum ciliare.

Adding uncertainty to any accounts of Buffel grass is that the genera Cenchrus and Pennisetum are not easily distinguishable, and caution should be taken to ensure that records of the species are credible (Pers. Comm. Helen Vonow, South Australian Herbarium).

Morphological and physiological differences between Buffel grass varieties have been studied on several occasions (GutierrezOzuna et al., 2009; Jacobs et al., 2004; Jorge et al., 2008; Mnif et al., 2005a; Morales-Romero and Molina-Freaner, 2008). These studies describe the range of dimensions to which the grass grows (Table 2.1).

A Pakistani study on the morpho-genetic variability between 20 Buffel grass accessions (Arshad et al., 2007) showed that most (38.7%) of the morphological variance between accessions used in the study were associated with the height of the plant, leaf area, number of leaves on the main tiller, part of internodes covered by leaf sheath, the number of branches per plant and the number of reproductive branches per plant (Arshad et al., 2007).

Intra-species variation has arisen both naturally and from the commercial development of new strains to improve productivity of pastoral land. Cultivars have been developed with increased growth rates, disease resistance and tolerance to a range of environmental conditions.

Consequently, knowledge about the suitability of various strains in different environments may be critical for effective control of infestations.
Generally, Buffel grass is apomictic (Bray, 1978), although rare sexual individuals have been identified (Akiyama et al., 2005). Seed spreads easily by wind, along water courses and human or animal traffic. Some varieties can also reproduce vegetatively through rhizomes and stolons.

The result of this is that a range of plant forms occur and can be observed growing in dense monotypic stands, as well as in small clumps or even lone tussocks throughout the landscape.

To thoroughly assess the threats posed by Buffel grass invasion, we must ask whether the invasive capacity also varies between sub-species. This has been studied in Mexico (Gutierrez-Ozuna et al., 2009) and Tunisia (Mnif et al., 2005b).

Both studies concluded that invasion success is not directly linked to genotypic variation, and that other factors such as phenotypic plasticity and propagule pressure could be major determinants of the invasion.

(Table 2.1) Approximate range of dimensions of Buffel grass morphology as described in the literature.

Trait	Dimensions

Plant height	20e150 cm
Stem thickness	1e3 mm
Leaf	1.5e30 cm long, 3e8 mm wide
Ligules	0.5e2 mm
Inflorescences	Yellowepurpleegrey
Time to flowering	Approximately 3 months from germination
Roots	Up to 2.4 m deep

2.4 Description

C. ciliaris is a fast-growing, shortly stoloniferous perennial that can flower in its first year of growth. Individual plants develop as clumps usually with only limited lateral spread, but a clump may eventually grow to >1 m in diameter.

Morphology (especially size) is highly variable depending on the genotype and the environment. Height of flowering culms may range from 15 cm to ~1.5 m. The width of the blades typically varies from 0.5-1.5 cm.

Inflorescences are bristly, typically from 3-15 cm long and 1-2 cm wide, and can range in color from tan to purple-tinged. Spikelets are 3-6 mm long with 2 florets, in clusters of 2-4, each surrounded by an involucre of feather-like bristles joined at the base, up to 15 mm long. Cariopses are ovoid, 1.5-2 mm long. Seed production is generally high, with most fascicles (detachable dispersal units) containing 1-2 viable, apomictic seeds.

2.5 Habitat

C. ciliaris grows in a variety of arid and semi-arid habitats, in particular those subject to disturbance. For further information see the 'Habitats' table and the

'Environmental requirements' paragraphs in the 'Biology and Ecology' text section.

(Table 2.2) Habitat List.

Category	Habitat	Presence	Status
Terrestrial-managed	Cultivated / agricultural land	Secondary/tolerated habitat	Harmful (pest or invasive)
	Disturbed areas	Secondary/tolerated habitat	Harmful (pest or invasive)
	Managed grasslands (grazing systems)	Principal habitat	Harmful (pest or invasive)
	Managed grasslands (grazing systems)	Principal habitat	Natural
	Managed grasslands (grazing systems)	Principal habitat	Productive/non-natural
	Rail / roadsides	Secondary/tolerated habitat	Harmful (pest or invasive)
	Urban / peri-urban areas	Secondary/tolerated habitat	Natural

2.6 Hosts/Species Affected

In its native range, *C. ciliaris* is reported as a weed of various crops including chickpea *Cicer arietinum* (Marwat et al., 2004), cotton (Rajput et al., 2008), potato (Shedayi et al., 2011) and maize *Zea mays* (Zair Muhammad, 2011), but specific impacts of the species on crops have not been quantified.

2.7 Biology and Ecology

2.7.1 Genetics

C. ciliaris is apomictic, meaning that it can produce asexual seeds that are genetically identical to the mother plant; however, genetic markers indicate that sexual reproduction is also likely (Kharrat-Souissi et al. 2011).

Tremendous genetic variation has been identified among *C. ciliaris* stocks; this results in variable morphology and ecological tolerances (Pengelly et al., 1992; Hignight et al., 1991).
Among 568 screened accessions, 54% were tetraploids with 36 chromosomes, 24% were pentaploids with 45 chromosomes, 4% were hexaploids with 54

chromosomes, >1% were septaploids with 63 chromosomes, and 17% were aneuploids (Burson et al., 2012).

Natural hybridization between *C. ciliaris* and *Pennisetum* species seems unlikely due to genetic incompatibilities (e.g. Marchais and Tostain 1997), but hybridization with *Cenchrus setigerus* has been reported, and hybridization between different introduced *C. ciliaris* varieties could lead to increased spread and invasiveness.

2.7.2 Reproductive Biology

Although limited spread of *C. ciliaris* may be observed at arid sites in dry years, unusually wet years can promote sudden and expansive seedling establishment across large areas (e.g. Burgess et al., 1991).

Once established, the plants are long-lived and highly tolerant of drought. *C. ciliaris* can potentially flower in its first year of growth (facultative annual), but established plants may live for more than a decade.

Seeds can remain viable in soil for at least four years (Winkworth 1971). Fresh seeds have low germination rates due to dormancy, which is typically lost over 4-16 weeks (Hacker and Ratcliff, 1989).

Seeds may germinate throughout the year in response to rain. Seedlings can tolerate ~60% shade (Pyon et al., 1977), but flowering and seed production might require more light.

2.7.3 Physiology and Phenology

C. ciliaris utilizes the C4 photosynthetic pathway, which provides an advantage under hot and dry growing conditions. During extended dry periods, above-ground plant parts may die back completely, but the rhizomes/root mass can survive and resprout quickly in response to rain. In seasonally dry environments, flowering is in response to adequate rain and is not strongly dependent on photoperiod.

In managed pastures, fire is often prescribed at 1-3 year intervals to maximize productivity and nutritional content of *C. ciliaris* as a forage and to control establishment of woody plants (Hamilton and Scifres 1982; Mayeux and Hamilton, 1983; Gupta and Trivedi, 2001).

2.7.4 Associations

C. ciliaris is commonly associated with scattered woody legumes such as *Prosopis* spp., *Acacia* spp. or *Leucaena leucocephala*. Other drought-tolerant grasses may co-occur with it including *Urochloa maxima* [*Panicum maximum*], *Setaria verticillata*, *Chloris* spp., *Hyparrhenia* spp. and *Eragrostis* spp. In South Africa, a *Cenchrus ciliaris* – *Cyperus marginatus* community has been described along seasonally dry riverbanks and drainage routes (Werger and Coetzee, 1977).

2.8 Environmental Requirements

C. ciliaris prefers dry or seasonally dry environments. Rao et al. (1996) reported optimal yields with 180-250 mm of rainfall concentrated in the growing season, although yields were also high with total rainfall of ~650 mm. There is substantial variation among varieties/genotypes in drought tolerance (Kharrat-Souissi et al., 2012).

C. ciliaris has been reported to persist on sandy (fast-draining) soils with annual rainfall as high as 1200 mm (Cox et al 1988). At higher rainfall, it may be outcompeted by taller grasses, or it may be increasingly susceptible to fungal pathogens such as *Magnaporthe* (*Pyricularia*) *grisea* (Perrott and Chakraborty, 1999).

In North America, Ibarra-F et al. (1995) found that *C. ciliaris* plantings failed to persist above 600 mm of annual rainfall, but only one cultivar was studied (T-4464, common buffel grass). *C. ciliaris* has spread successfully from planting in areas having a dry period of 150-210 days; mean maximum temperature ranged between 24 and 32°C and minimum temperatures in the coldest month ranged from 5 to 15°C (Ibarra-F. et al. 1995).

Common buffel grass has low tolerance of freezing temperatures, but can tolerate very high air temperatures (approaching 45°C) (Barrera and Castellanos, 2007). *C. ciliaris* does not tolerate water-logged soils.

In pot experiments, Anderson (1974) found that *C. ciliaris* plants were often killed when exposed to 15-20 days of artificial flooding, but there was substantial variation among cultivars. *C. ciliaris* is reportedly intolerant of excessive manganese, and this might be related to low availability of calcium under such conditions (Smith, 1979)

There is great potential for breeding *C. ciliaris* that can tolerate different environmental conditions such as heavy clay soils (Hacker and Waite 2001), higher salinity (Griffa et al., 2010) or cold environments (Hussey and Burson, 2005).

2.9 Climate

(Table 2.3) Climate

Climate	Status	Description	Remark
A - Tropical/Megathermal climate	Preferred	Average temp. of coolest month > 18°C, > 1500mm precipitation annually	
As - Tropical savanna climate with dry summer	Preferred	< 60mm precipitation driest month (in summer) and < (100 - [total annual precipitation{mm}/25])	
Aw - Tropical wet and dry savanna climate	Tolerated	< 60mm precipitation driest month (in winter) and < (100 - [total annual precipitation{mm}/25])	
BW - Desert climate	Tolerated	< 430mm annual precipitation	
Cs - Warm temperate climate with dry summer	Tolerated	Warm average temp. > 10°C, Cold average temp. > 0°C, dry summers	

Parameter	Lower limit	Upper limit	Description
Dry season duration	2	9	number of consecutive months with <40 mm rainfall
Mean annual rainfall	375	1200	mm; lower/upper limits

(Table 2.4) Rainfall

Rainfall Regime

- Bimodal
- Summer
- Winter

2.10 Soil Tolerances

Soil drainage

- free

Soil reaction

- alkaline
- neutral

Soil texture

- heavy
- light
- medium

Special soil tolerances

- infertile
- saline
- shallow

2.11 Natural enemies

Natural enemy	Type	Life stages	Specificity	References	Biological control in	Biological control on
Aeneolamia albofasciata	Herbivore	Whole plant	not specific			
Magnaporthe oryzae	Pathogen	Whole plant	not specific			
Mampava rhodoneura	Herbivore	Seeds	not specific			

(Table 2.5) Natural Enemies

Notes on Natural Enemies

Magnaporthe (*Pyricularia*) *grisea* is a blast fungus that has caused dieback of *C. ciliaris* in the United States and Mexico, where the widely planted common buffel grass (cultivar T-4464) appears to be especially susceptible (Rodriguez et al., 1999). Other cultivars have shown tolerance to the fungus (Díaz-Franco and Méndez-Rodríguez, 2005). In Australia, dieback of *C. ciliaris* that seems to be associated with an unidentified pathogen has been reported (Makiela and Harrower, 2008).

Injury and death of *C. ciliaris* due to feeding by spittle bugs (*Aeneolamia albofasciata* Lall.) has been observed in Mexico, especially in wet years (Martin-R et al., 1995). Around Queensland, Australia, larvae of the buffel grass seed moth (*Mampava rhodoneura* Turner) sometimes feed on seed heads (Cook et al., 2005), but they do not appear to be economically important.

2.12 Means of Movement and Dispersal

C. ciliaris has been deliberately transported between most arid and semi-arid regions of the world. After deliberate plantings, seeds are spread by wind, water, machinery and animals (including potential for attachment to human clothing).
Natural Dispersal (non-biotic)

Wind may be the primary mode of dispersal over short-to-moderate distances. Common establishment of *C. ciliaris* along drainage areas (Puckey et al. 2007) suggests water dispersal in some cases.

Vector Transmission (Biotic)

One study found that seeds ingested by cattle remained intact but failed to germinate (Gardener et al ., 1993), suggesting that internal dispersal by animals may be limited; however, seed-bearing fascicles have bristles that can cling externally to animals and clothing, and seeds are small (1.5-2 mm) and could be dispersed with mud on hooves or vehicles.
Ants can also disperse *C. ciliaris* seeds (Goldsmith et al., 2008).

Accidental Introduction

Roadside gusts associated with passing vehicles are likely to facilitate the spread of seeds along roadsides (and similarly, along railways). Roadside mowing could also facilitate spread of seeds.

Intentional Introduction

C. ciliaris has been planted as a fodder and for erosion control in most warm arid and semi-arid regions of the world.

(Table 2.6) **Pathway Causes**

Cause	Notes	Long Distance	Local	References
Aid		Yes	Yes	Marshall et al., 2012
Breeding and propagation		Yes		Marshall et al., 2012
Crop production		Yes	Yes	Marshall et al., 2012
Disturbance			Yes	
Forage		Yes	Yes	Marshall et al., 2012
Habitat restoration and improvement		Yes	Yes	Warren and Aschmann, 1993
Hitchhiker		Yes	Yes	
Intentional release		Yes	Yes	Marshall et al., 2012
Research		Yes		Brenner, 2010

Vector	Notes	Long Distance	Local	References
Land vehicles		Yes	Yes	
Livestock			Yes	
Soil, sand and gravel		Yes	Yes	
Water			Yes	
Wind			Yes	

(Table 2.7) Pathway Vectors

2.13 Economic Impact

In its native range, *C. ciliaris* is reported as a weed of various crops including chickpea *Cicer arietinum* (Marwat et al., 2004), cotton (Rajput et al., 2008), potato (Shedayi et al., 2011) and maize *Zea mays* (Zair Muhammad, 2011), but economic costs have not been directly quantified.

It has been reported as a host for the sugarcane whitefly (*Neomaskellia bergii* (Signoret)), an important economic pest of sugarcane (Palmer, 2009a). It can also serve as a host for the rusty plum aphid (*Hysteroneura setariae* (Thomas)), which is a vector of several viruses that impact economic crops (Palmer, 2009b).

Substantial expenditures have been directed towards control of *C. ciliaris* in protected natural areas. One example is at Organ Pipe Cactus National Monument, Arizona, USA, where *C. ciliaris* was ranked as the top priority weed for eradication and more than 890 person-hours were dedicated to the effort (Rutman and Dickson, 2002). Money is being spent on numerous control and eradication programs around the world (e.g. Dixon et al, 2002; Daehler and Goergen, 2005).

2.14 Environmental Impact

2.14.1 Impact on Habitats

Erosion and dust control via ground covering can be a positive environmental impact of *C. ciliaris* (Warren and Aschmann, 1993; Carroll and Tucker, 2000). Dominance by the species in pastures can also reduce the abundance of unpalatable or otherwise noxious weeds, which could have positive environmental effects if fewer chemical herbicides are used in pastures.

C. ciliaris has reportedly become an important component of the diet of native wombats in parts of Australia (Low, 1997), and it is probably consumed to some extent by native herbivores in other parts of the world where it has been introduced (e.g. it is eaten by native deer in the southwestern United States (Everitt and Gonzalez 1979).

In the Sonoran region of North America, deforestation for the planting of *C. ciliaris* pasture has affected hundreds of thousands of hectares (Franklin et al., 2006). After initial planting, *C. ciliaris* invades cactus shrublands, forming a continuous understorey that promotes fire, which kills native plants and drives a grass-fire cycle that converts the system to a buffel grassland (e.g. see Fig 2 in Brenner, 2010).

Affected native species include creosote (*Larrea tridentata*), saltbush

(*Atriplex* spp.), bursage (*Ambrosia* spp.) and saguaro (*Carnegiea gigantea*). Dead biomass of *C. ciliaris* tends to accumulate across years in the absence of fire, leading to increased fire risk in unburned stands over time. *C. ciliaris* fires may be hotter and more intense than fires fuelled by native vegetation (McDonald and McPherson, 2011).

Buffel grasslands have also been reported to have lower productivity than some native desert shrublands (Franklin et al., 2006). In the south-eastern range of the species in Mexico, soil beneath *C. ciliaris* that had been established for 10-20 years had 40% lower nitrogen than neighbouring uninvaded areas, and this pattern could suggest declines in productivity over time following invasion (Ibarra-Flores et al., 1999).

Arriaga et al (2004) used a model to project the future distribution of the species in Mexico, and concluded that desert scrub, mesquite woodland, and tropical deciduous forest were at the greatest risk of invasion. In Australia, similar conversion of native *Eucalyptus* woodland and *Acacia* woodland to buffel grassland occurs via a grass-fire cycle (Butler and Fairfax, 2003; Miller et al., 2010).

2.14.2 Impact on Biodiversity

In Australia, Fairfax and Fensham (2000) found that plant species diversity was significantly lower in *C. ciliaris* pastures as compared to native grass pastures. Franks (2002) surveyed remnant *Eucalyptus* woodland in Australia and found that *C. ciliaris* cover was negatively correlated with understorey native plant diversity.

In semi-arid vegetation of central Australia, invaded areas had reduced plant richness, and native plant population declines were observed among all plant growth forms (Clarke et al., 2005). Jackson (2005) found that diversity of native herbaceous species declined with increasing *C. ciliaris* cover in open woodlands in Australia.

Daehler and Carinio (1998) found that *C. ciliaris* invasion was associated with substantially reduced native and non-native plant richness in Hawaii. Various endangered and vulnerable endemic plants are threatened by *C. ciliaris* invasion in Australia and the United States.

The possibility of allelopathy by *C. ciliaris* is supported by laboratory assays in which *C. ciliaris* leachates reduced germination of other species (Fulbright and Fulbright 1990; Farrukh Hussain et al., 2011), but these findings need to be confirmed in field studies.

In a study in Texas, USA, Flanders et al. (2006) compared bird use of native-dominated vegetation versus areas invaded by *C. ciliaris* and Lehmann lovegrass (*Eragrostis lehmanniana*). Although shrub and canopy cover were similar at native and invaded sites, abundance of many birds was greater in native vegetation, particularly among birds that forage on the ground.

Arthropod abundance was also greater in the native vegetation, which may partly explain the greater abundance of birds there. In a semi-arid woodland in Australia, *C. ciliaris* invasion appears to have altered the distribution of native reptiles (Eyre et al., 2009).

In the United States and Mexico, reduced woody cover and plant litter associated with *C. ciliaris* invasion and fire have likely reduced the area of suitable habitat for desert tortoises (*Gopherus agassizii*), jaguarundis (*Felis yaguarondi*), and ocelots (*Felis pardalis*) (Esque et al. 2007).

In some cases, rare animal species are threatened by *C. ciliaris* invasion. One example is Slater's skink (*Egernia slateri*) in Australia, where "degradation of its alluvial habitat as a result of invasion by the introduced buffel grass and associated changes in fire regimes appears the most likely causes of the species' decline" (Pavey, 2006).

Another example is the Desert Sand Skipper (*Croitana aestiva*), an endangered butterfly restricted to a small area in central Australia, most of which has been invaded by *C. ciliaris* (Brabey et al., 2006). The larval host plant of this butterfly is unknown, but other species in the genus feed on native grasses, which have greatly declined following *C. ciliaris* invasion.

Also in Australia, *C. ciliaris* has been identified as a threat to the endangered bridled nailtail wallaby because "buffel grass has out-competed other herbaceous species, which are the preferred browse of wallabies. In addition, dense swards of buffel grass may act as a barrier to wallaby dispersal" (Lundie-Jenkins and Lowry, 2005, p. 27).

Because of the tremendous spatial extent of *C. ciliaris* invasions and the severe modification of habitat associated with these invasions (including altered fire regime), it is likely that habitats for numerous rare species (both plants and animals) have been completely subsumed by *C. ciliaris*.

(Table 2.8) Threatened Species

Threatened Species	Conservation Status	Where Threatened	Mechanism	References	Notes
Ayenia limitaris (border ayenia)	EN (IUCN red list: Endangered) EN (IUCN red list: Endangered)		Competition - monopolizing resources	Esque et al., 2007	
Croitana aestiva (desert sand skipper)	EN (IUCN red list: Endangered) EN (IUCN red list: Endangered)			Braby et al., 2006	
Egernia slateri (Slater's skink)	EN (IUCN red list: Endangered) EN (IUCN red list: Endangered)			Pavey, 2006	
Manihot walkerae (Walker's manioc)	EN (IUCN red list: Endangered) EN (IUCN red list: Endangered)		Competition - monopolizing resources	Esque et al., 2007	
Minuria tridens (Minnie daisy)	VU (IUCN red list: Vulnerable) VU (IUCN red list: Vulnerable)	Australian Northern Territory	Competition - monopolizing resources	Nano and Pavey, 2008	
Olearia macdonnellensis	VU (IUCN red list: Vulnerable) VU (IUCN red list: Vulnerable)	Australian Northern Territory	Competition - monopolizing resources	Nano and Pavey, 2008	
Onychogalea fraenata (bridled nailtail wallaby)	EN (IUCN red list: Endangered) EN (IUCN red list: Endangered)	Queensland		Lundie-Jenkins and Lowry, 2005	
Physaria thamnophila (Zapata bladderpod)	EN (IUCN red list: Endangered) EN (IUCN red list: Endangered)		Competition - monopolizing resources	Esque et al., 2007	

2.15 Social Impact

C. ciliaris invasion in Australia is apparently leading to declines in some traditional Aboriginal food plants such as spinifex (*Triodia* sp.) (Low, 1997). In its native range, *C. ciliaris* stands harbour ticks that carry human and wildlife diseases (Wanzala and Okanga, 1996), and it is conceivable that humans and animals walking in areas invaded by the species will be more susceptible to tick bites, as compared to the more open groundcover that existed prior to invasion. Fires promoted by *C. ciliaris* can threaten homes and other structures and facilities utilized by people.

2.16 Risk and Impact Factors

Invasiveness

- Proved invasive outside its native range
- Has a broad native range
- Abundant in its native range
- Highly adaptable to different environments
- Is a habitat generalist
- Tolerates, or benefits from, cultivation, browsing pressure, mutilation, fire etc.

- Pioneering in disturbed areas
- Long live
- Fast growing
- Has high reproductive potential
- Has propagules that can remain viable for more than one year
- Reproduces asexually
- Has high genetic variability

Impact outcomes

- Damaged ecosystem services
- Ecosystem change/ habitat alteration
- Modification of fire regime
- Modification of successional patterns
- Monoculture formation
- Reduced native biodiversity
- Threat to/ loss of native species

Impact mechanisms

- Competition - monopolizing resources
- Rapid growth

Likelihood of entry/control

- Highly likely to be transported internationally deliberately

2.17 Uses

2.17.1 Economic Value

C. ciliaris has had positive economic impacts as a fodder for domestic animals. Its drought tolerance and tolerance of overgrazing allow increased production compared to native grasslands, especially in marginal environments (e.g. Martin-R et al. 1995).

It is particularly valued for its ability to produce grazeable biomass quickly in response to periodic or unpredictable rain (Marshall et al., 2011). However, some studies have reported declines in productivity of *C. ciliaris* pastures over time, presumably due to nutrient limitations (Ibarra-Flores et al., 1999),

so a long-term maintenance of economic benefits is uncertain. *C. ciliaris* has been used to restore productivity to degraded land around mining sites in Australia (Bisrat et al, 2004), yielding economic benefits. In India, its seeds are sometimes mixed with millet for bread-making (Quattrocchi 2006), but the economic value of human-consumed seeds is probably small.

2.17.2 Uses List

Animal feed, fodder, forage

- Forage

Environmental

- Erosion control or dune stabilization
- Land reclamation
- Landscape improvement
- Revegetation
- Soil conservation
- Soil improvement

Diagnosis

No information is available on specific tests used for detection; see the section on 'Similarities to Other Species' for a comparison of *C. ciliaris* to similar-looking species.

2.18 Similarities to Other Species/Conditions

Cenchrus setiger Vahl (*C. setigerus*, birdwood grass) and *Cenchrus pennisetiformis* Hochst. & Steud. (Cloncurry buffel grass) have been used less frequently as tropical pasture grasses. *C. setigerus* has spikelet bristles that are barely emergent from the spikelet (2-4 mm), while *C. ciliaris* has bristles that are obviously emergent from the spikelet (6-18 mm long), with one bristle that is longer than the rest.

C. pennisetiformis has long bristles like *C. ciliaris*, but the bristles are fused at the base to a greater degree (1-2.5 mm) as compared to *C. ciliaris* (0.5-1 mm) (Clayton et al., 2012). *C. pennisetiformis* might be a natural hybrid between *C. ciliaris* and *C. setigerus* (Cook et al. 2005). *Cenchrus echinatus* L. (southern sandspur) is a noxious weed with stiff spikelet bristles that can puncture human skin, whereas *C. ciliaris* bristles will bend or break if spikelets are squeezed between two fingers.

Cenchrus spp. differ from *Pennisetum* spp. in that the involucre of bristles is fused at the base in *Cenchrus*, while in *Pennisetum* the bristles are individually free.

2.19 Prevention and Control

C. ciliaris seedling establishment is especially prolific in gaps in vegetation (McIvor 2003), so disturbed or naturally open environments are at the highest risk of initial and rapid invasion. Road or building construction in or near natural areas can allow an opportunity for rapid invasion (e.g. Barrera, 2008), so sites of disturbance should be carefully monitored.

C. ciliaris spreads rapidly along roadsides, and from there it spreads into a wide range of open or semi-open environments with the assistance of fire or other disturbances. Isolated roadside infestations should be a priority for control in an effort to slow spread and reduce risk of invasion of surrounding habitats.

Puckey et al. (2007) found that early detection and mapping within natural areas was most efficiently done by aerial (helicopter) surveys, while also considering probabilistic factors such as proximity to rail tracks or drainage sites, which increases efficiency of searches and detection.

In areas where *C. ciliaris* is promoted for pastures, it might be possible to use a buffer-zone management approach to keep it from spreading into conservation areas (Friedel et al., 2011). A buffer zone approach would be most feasible for protecting relatively small, high-value natural areas.

2.19.1 Physical/Mechanical Control

Once *C. ciliaris* has become established in an area, manual removal (by digging) has sometimes been effective for small areas (e.g. Daehler and Goergen 2005), but manual removal is very labour intensive and plants can grow back from small rhizome pieces left in the soil.

Large numbers of volunteers have been used to pull and dig out *C. ciliaris* spreading along roadsides (e.g. Southern Arizona Buffelgrass Coordination Center, http://www.buffelgrass.org – see Links to Websites).

The excavated material is collected in plastic bags and removed from the site.

Mowing is not effective in reducing *C. ciliaris* invasion, and may help disperse the species if the mown plants have inflorescences. On the other hand, mowing or grazing may be helpful in reducing fire risks as part of an integrated management plan.

2.19.2 Biological Control

Classical biocontrol is unlikely to be an option because of actual and/or perceived benefits of *C. ciliaris*, and also because classical biocontrol of grasses has generally proven difficult. However, use of grazers to manage the species could be an option for reducing its dominance (and fuel load) in some areas (Friedel et al., 2011).

2.19.3 Chemical Control

Dixon et al. (2002) found that glyphosate effectively killed *C. ciliaris* and haloxyfop was successfully used to kill seedlings and as a grass-specific pre-emergent herbicide to prevent recolonization. Daehler and Goergen (2005) also found that glyphosate effectively killed established *C. ciliaris*.

At higher concentrations, fluazifop-p, a grass-specific herbicide, was also effective at killing *C. ciliaris* (Dixon et al., 2002). In all cases, plants need to be actively growing at the time of initial herbicide application. Dixon et al. (2002) found that an ideal timing for herbicide application exists a few weeks after rain, when established plants are actively growing and new seedlings have also germinated.

Control rates were approximately 98% across an area of several hectares. Control sites need to be monitored for several years following seasonal rains, as seedlings will continue to emerge from the seed bank.

2.20 Ecosystem Restoration

After control of dense *C. ciliaris* populations, revegetation with desired plants is important to minimize re-invasion risks and to reduce erosion. Naturally open areas pose a unique challenge because after *C. ciliaris* has been removed, the natural biological soil crust has been disrupted and it may require a decade or more to redevelop (Belnap and Eldridge 2001), leaving the area susceptible to reinvasion and/or soil erosion.

2.21 Distribution Maps

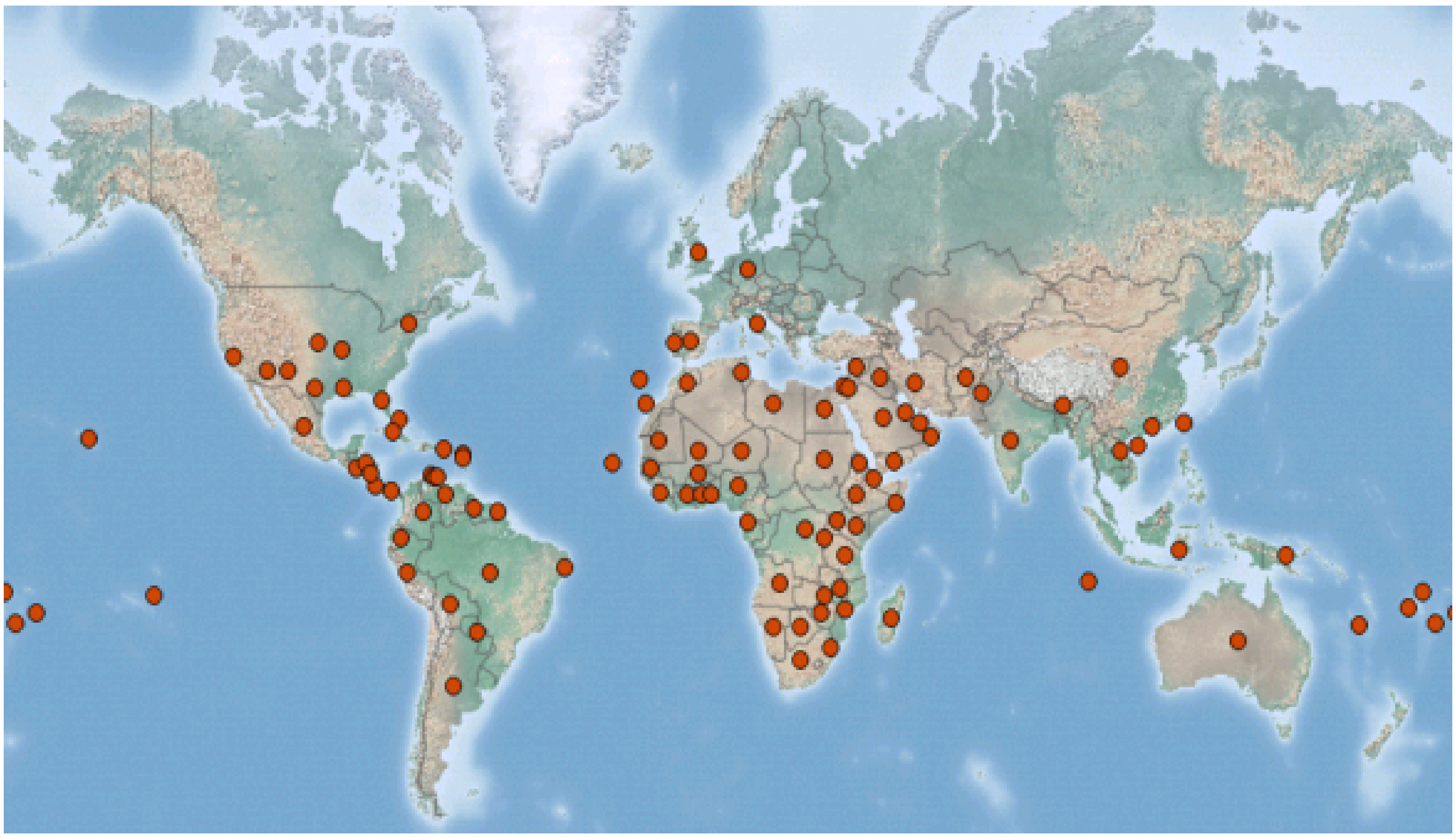

Fig 2.1 Location spread plant in the world

Chaptre3

Material & method

1 Sampling and sample preparation

3.1 Plant& wastewater sample location:

The plant taken from faculty garden from surface soil

The wastewater taken from Dolphin factory in the General Investment area

Fig 3.1 Plant sample location

Fig 3.2 Wastewater sample location

3.2 Material &method

This work aims to treatment of waste water using plant (*Cenchrus ciliaris*) Physicochemical parameters of wastewater samples (PH, DO, COND, SALINTY) will be analyzed.

3.2.1Plant preparation

3.3.1.1 Plant extract *Cenchrus ciliaris*:-

1- Drying of plant leaves

2- Grinding of leaves

3-Wight the grinded leaves

4-weigh 5 gm of plant leaves "as organic acid is present in the leaves"
5-put this weigh in 150ml of water
6- Put the sample in a water bath on the heater at 45 degrees for 24 hours

7-filter the sample by a filter paper

Treatment of wastewater using plant extract (*Cenchrus ciliaris*)
Take a sample 50 ml of wastewater from a textile or chemicals factory
Add different concentration of plant extract (*Cenchrus ciliaris)* on the wastewater sample

Table3.1: Physical & chemical measurement before and after using Cenchrus ciliaris by using plant extract 10% at T=28 C

10% of Plant extract *Cenchrus ciliaris*
-20% of Plant extract *Cenchrus ciliaris*
-30% of Plant extract *Cenchrus ciliaris*
- wait for 7 days and measure (PH, DO, COND, and SALINTY) (Serag, 2018)

Fig 3.3 leaves of plant

Fig 3.4 Sample preparation

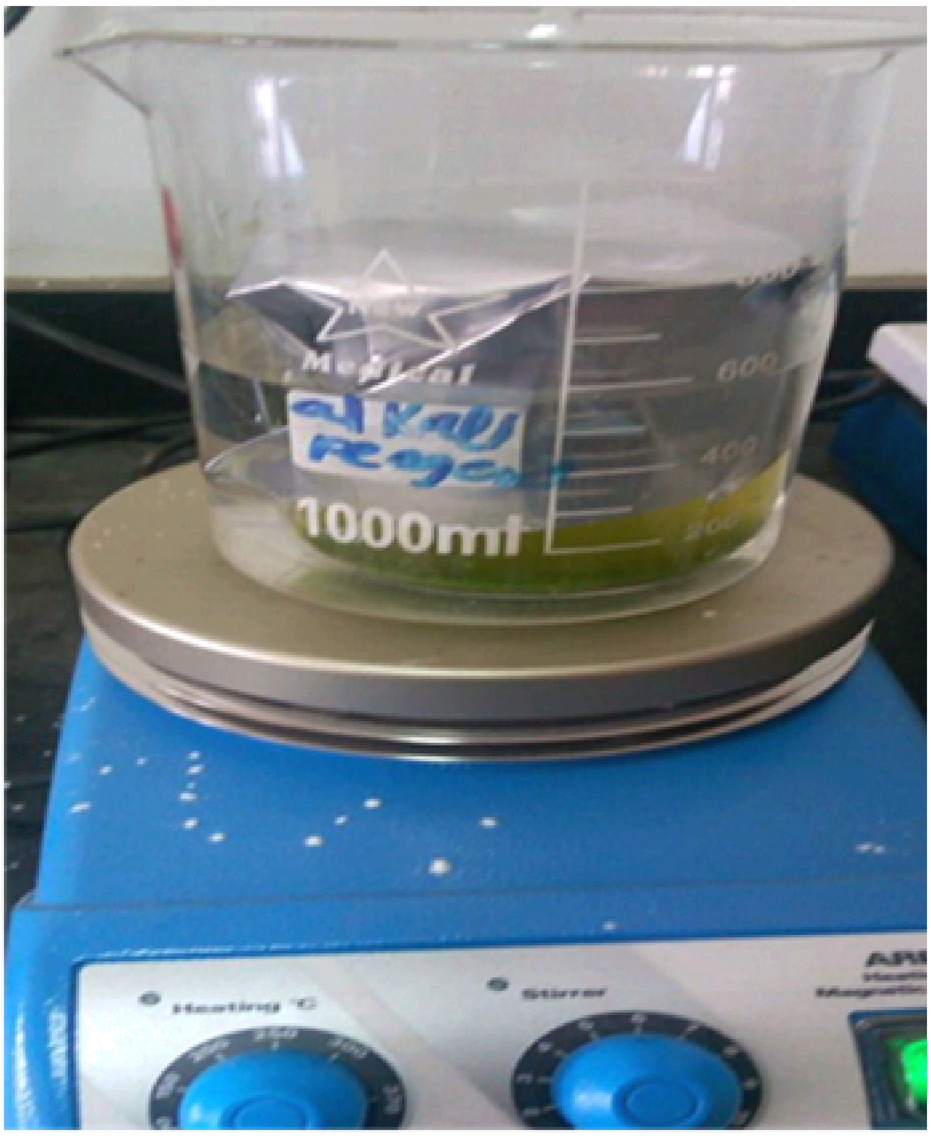

Fig 3.5 Sample on heater

Fig 3.6 sample after 7 days

3.2.2Wastewater analysis

Physicochemical parameters of wastewater samples (PH, DO, COND, SALINTY) will be analyzed using

Part 2

3.3.2.1 Dry plant (Cenchrus ciliaris)

1-collection of plant

2-use plant (leaves/ spikes/ branched)

3-dry the plant on temperatures70C on the oven

4-grending of plant parts (leaves, spikes, branched)

5-add different weigh from plant

- 1&2gm of leaves
- 1&2gm of spikes
- 1&2gm of branched

6- Put different weigh plant to wastewater sample

7-measure wastewater sample without addition of plant (blank)

8 -Wait for 7 days and measure (PH, DO, COND, SALINTY

(Serag, 2018)

Fig 3.7 sample after drying

Fig 3.8 sample Grinding

Fig 3.9 sample filtration

Fig 3.10 sample preparation

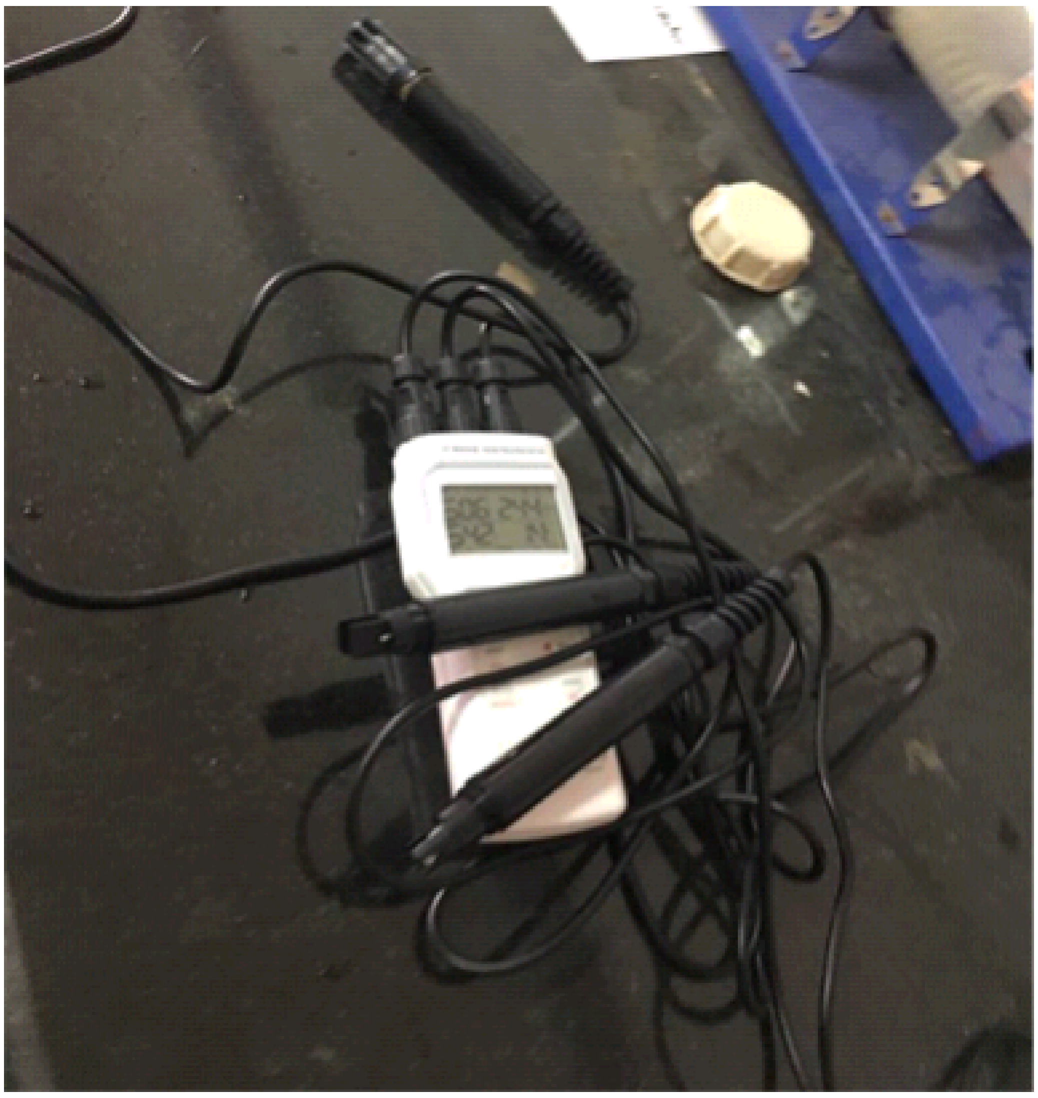

Fig3.11Malutimeter to using parameter of water

4.3 Discussion

Improving physical and chemical properties (PH, Cond, Salinity and DO), change sample's color and precipitate stain in the sample

Chapter 4
Result& discussion

Parameter	Ph	Cond	DO	Salinity
Control	6.7	1267	71.3%	0.62
sample	8	1500	70%	0.74

4 Result of experiment part1

Table4.1: Physical & chemical measurement before and after using Cenchrus ciliaris by using plant extract 10% at T=28 C

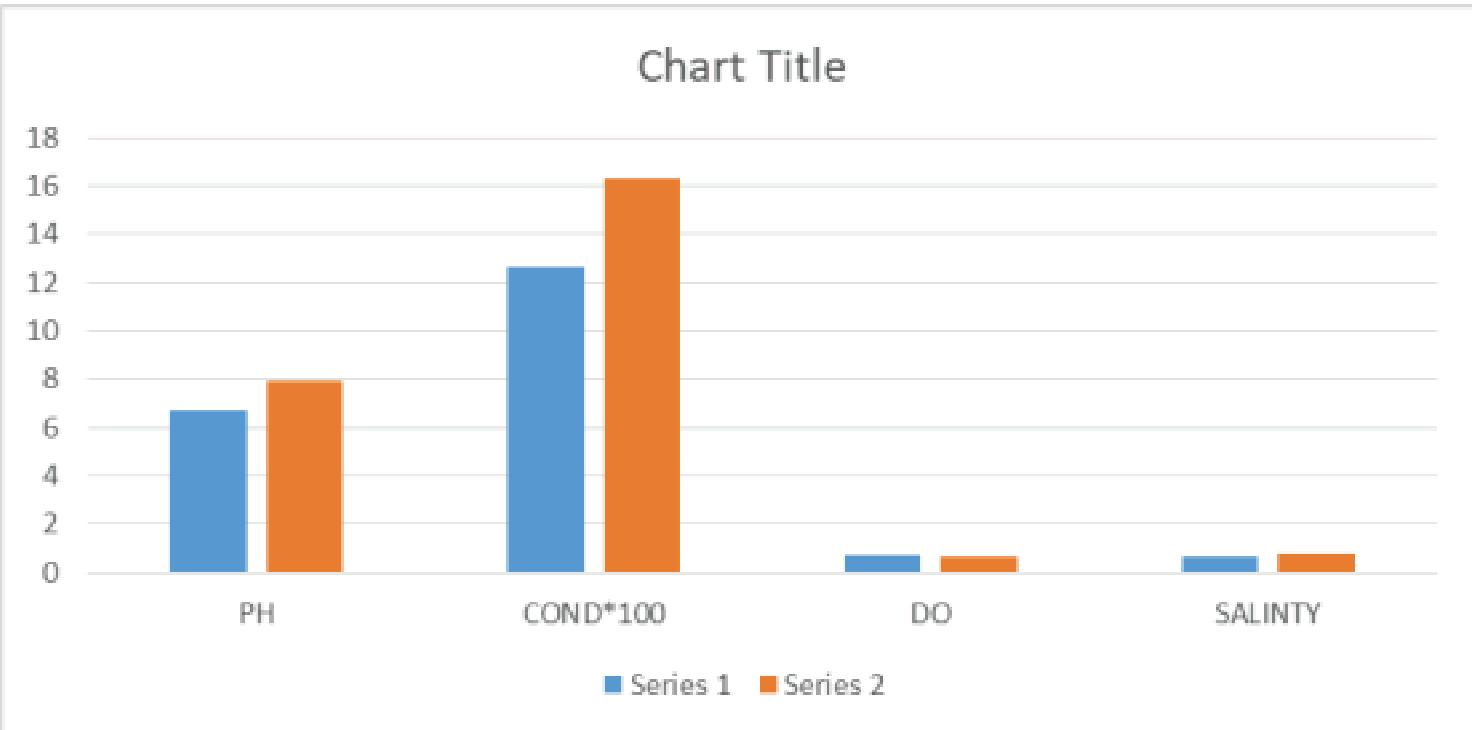

Fig 4.1: Physical & chemical measurement before and after using Cenchrus ciliaris by using plant extract 10% at T=28 C

Table 4.2: Physical & chemical measurement before and after using Cenchrus ciliaris by using plant extract 20% at T=28 C

Parameter	Ph	Cond	DO	Salinity
Control	6.7	1267	71.3%	0.62
Sample	*7.9*	*1630*	*65%*	*0.8*

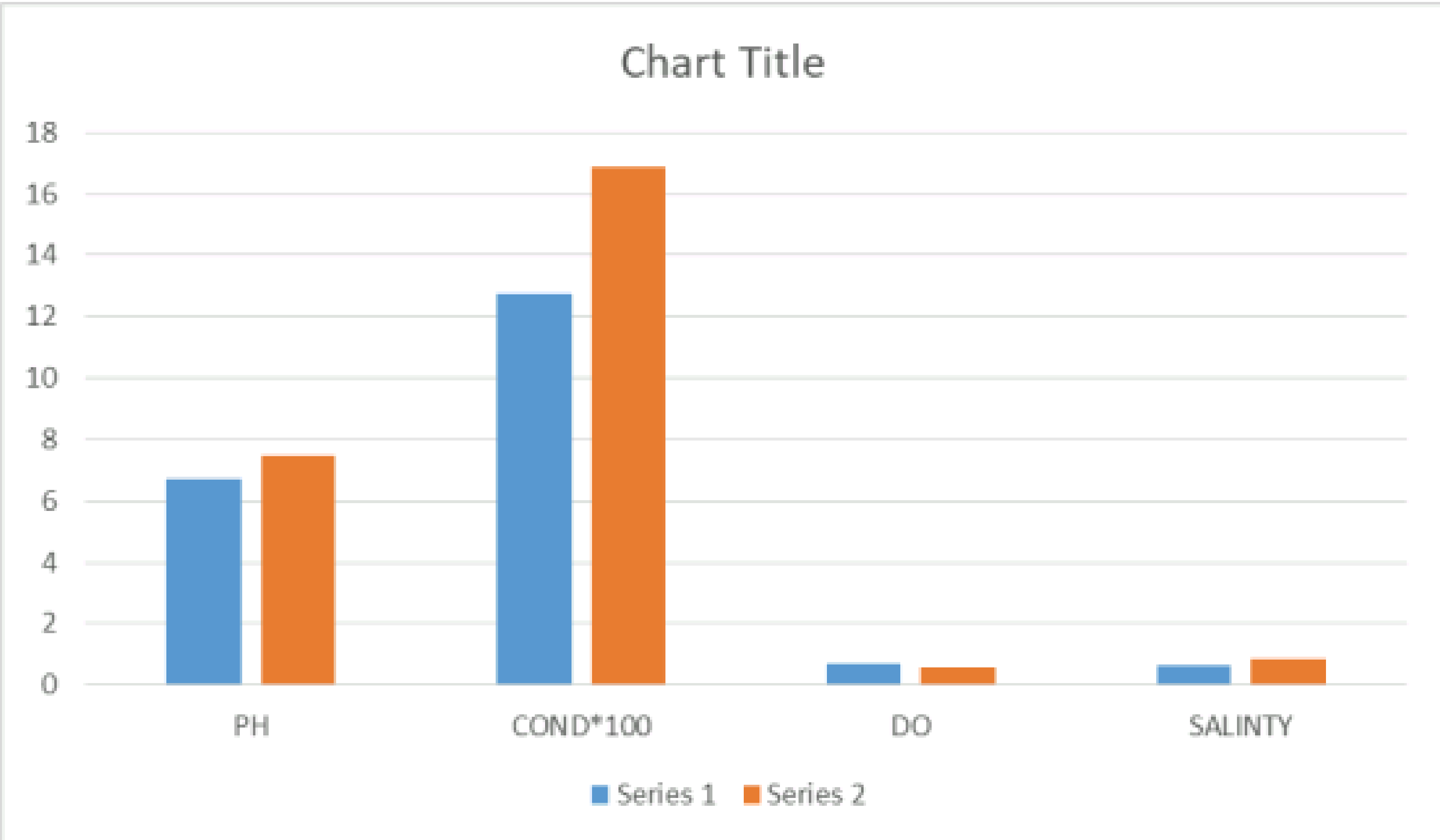

Fig 4.2 Physical & chemical measurement before and after using Cenchrus ciliaris by using plant extract 20% at T=28 C

Table 4.3: Physical & chemical measurement before and after using Cenchrus ciliaris by using plant extract 30% at T=28C

Parameter	Ph	Cond	DO	Salinity
Control	6.7	1267	71.3%	0.62

Sample	*7.5*	*1690*	*60%*	*0.85*

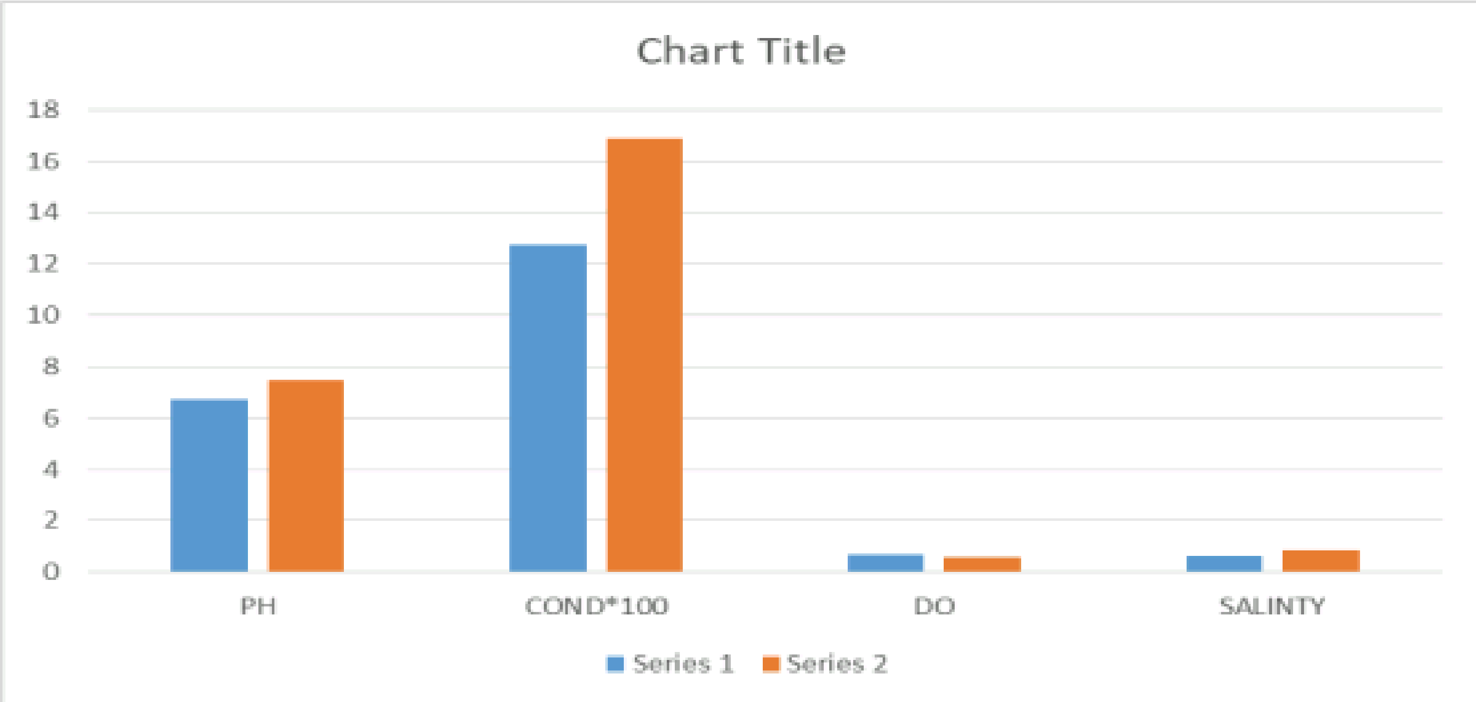

Fig 4.3 4.3: Physical & chemical measurement before and after using Cenchrus ciliaris by using plant extract 30% at T=28C

4 Result of experiment part 2

Table 4.4: Physical & chemical measurement before and after using Cenchrus ciliaris by using leaves of plant 1gm/50ml at T=27.5C

Parameter	Ph	Cond	DO	Salinity
Control	6.7	1267	71.3%	0.62
sample	6.9	2.7	92%	1.35

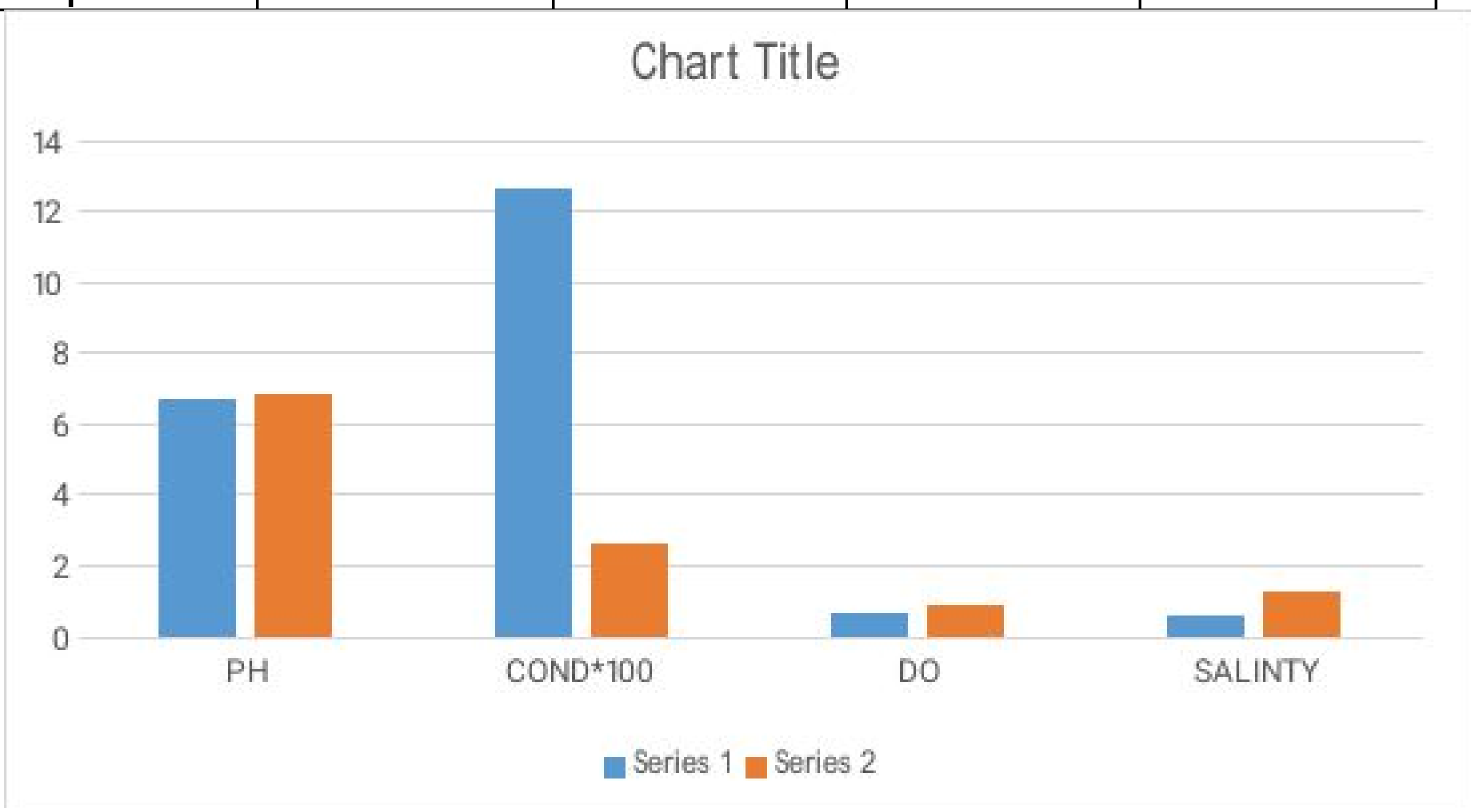

Fig 4.4: Physical & chemical measurement before and after using Cenchrus

ciliaris by using leaves of plant 1gm/50ml at T=27.5C

Table 4.5: Physical & chemical measurement before and after using *Cenchrus ciliaris* by using leaves of plant 2gm/50ml at T=27.5C

Parameter	Ph	Cond	DO	Salinity
Control	6.7	1267	71.3%	0.62
sample	5.6	3.79	60%	1.95

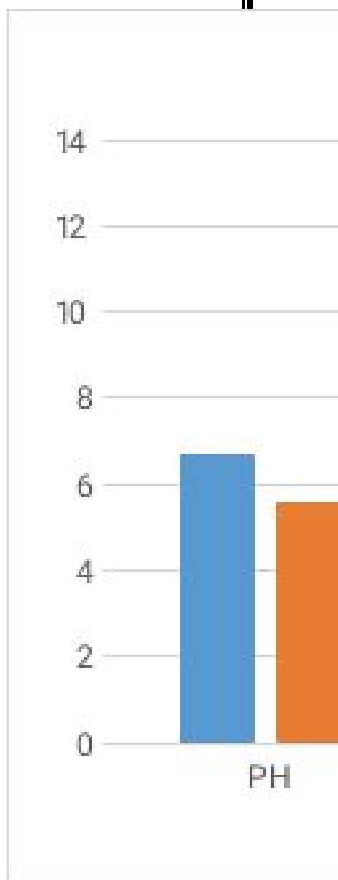

Fig 4.5: Physical & chemical measurement before and after using *Cenchrus ciliaris* by using leaves of plant 2gm/50ml at T=27.5C

Table 4.6: Physical & chemical measurement before and after using *Cenchrus ciliaris* by using branch of plant 1gm/50ml at T=28C

Parameter	Ph	Cond	DO	Salinity
Control	6.7	1267	71.3%	0.62
sample	7.1	3.40	80%	1.70

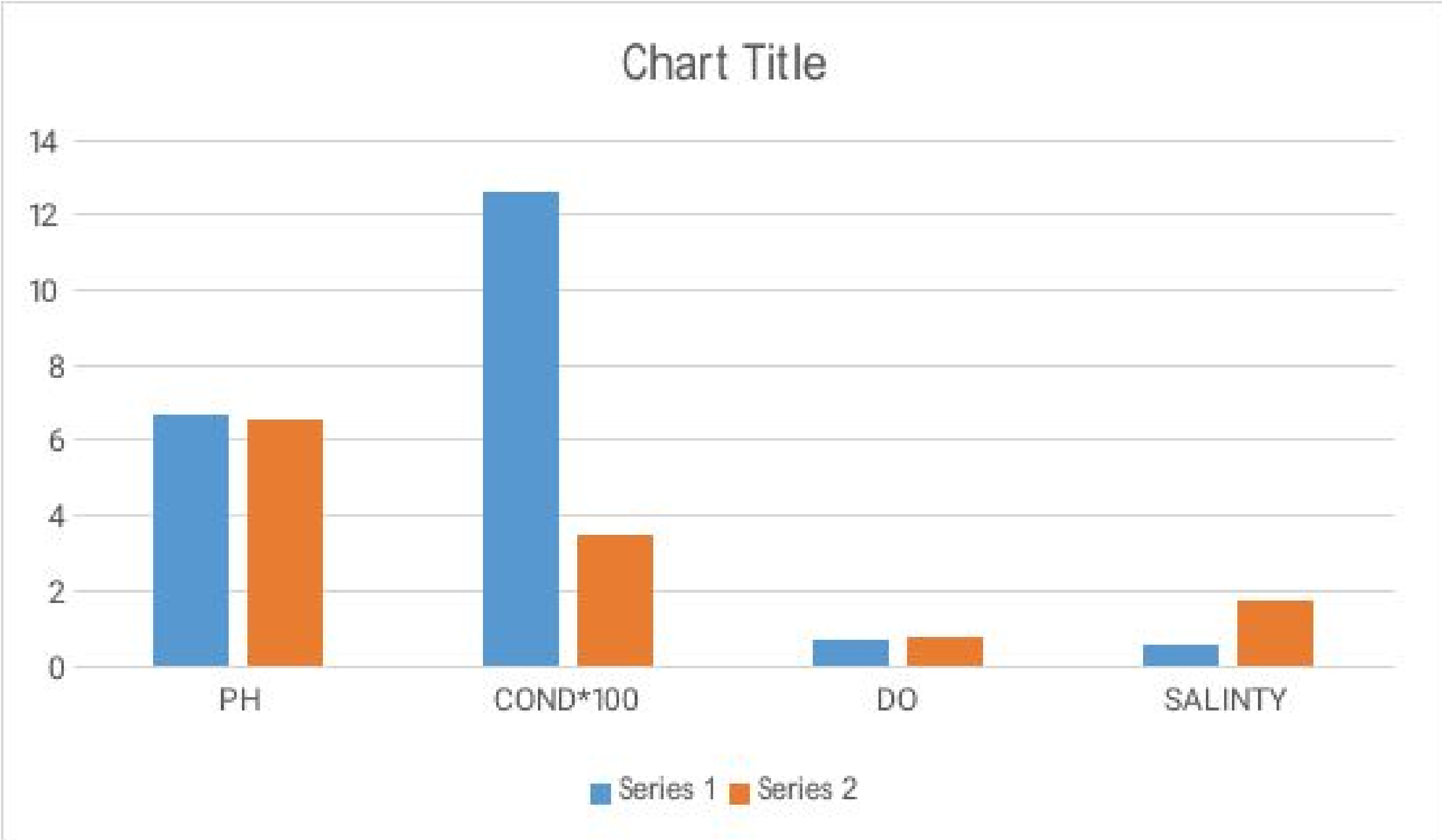

Fig 4.6 Physical & chemical measurement before and after using *Cenchrus ciliaris* by using branch of plant 1gm/50ml at T=28C

Table 4.7: Physical & chemical measurement before and after using *Cenchrus ciliaris* by using branch of plant 2gm/50ml at T=28C

Parameter	Ph	Cond	DO	Salinity
Control	6.7	1267	71.3%	0.62
sample	6.6	3.5	80%	1.80

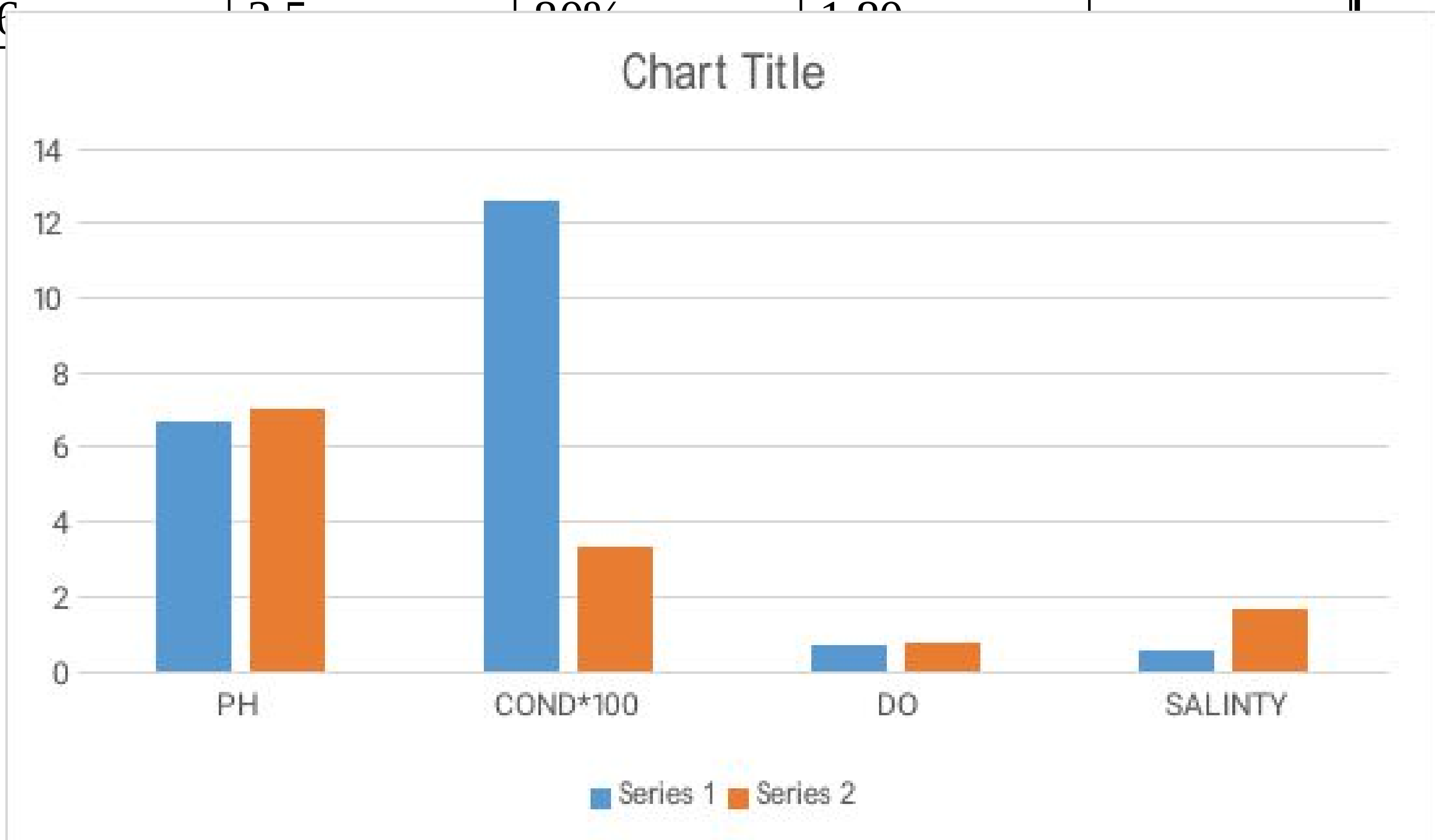

Fig 4.7: Physical & chemical measurement before and after using *Cenchrus ciliaris* by using branch of plant 2gm/50ml at T=28C

4.3 Discussion

Improving physical and chemical properties (PH, Cond, Salinity and DO), change sample's color and precipitate stain in the sample

Conclusion

- Industrial waste water discharged was contaminated with a wide range of toxic chemicals and pathogens causing serious health problems in human health and environment. Aquatic plants are very effective in removing heavy metals from polluted water. Plant assimilation of nutrients and its subsequent harvesting are another mechanism for pollutant removal. Low cost and easy maintenance make the aquatic plant system attractive to use. Eichhornia crassipes can also remove other toxins, such as cyanide, which is environmentally beneficial in areas that have endured gold mining operations.

Conclusion the Eichhornia plant reduce pH 10 to 8, TDS 4500 mg/L to 26 00 mg/L and other parameters also reduced 17-28%. Phytoremediation is suitable and low cost technology to remove degrade the pollution from industrial effluent like tannery effluent.The Eichhornia Crassipes is an extraordinary tool for effluent treatment if it is properly concentrated on Phytoremediation technology. It could be utilized the benefits and safe of our environment

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